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(54) **CAMERA MODULE**

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G03B 13/36 (2013.01); **H04N 23/52**

(2023.01); **H04N 23/67** (2023.01); **H04N**

23/6812 (2023.01); **H04N 23/687** (2023.01);

G02B 7/021 (2013.01)

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See application file for complete search history.

(56)

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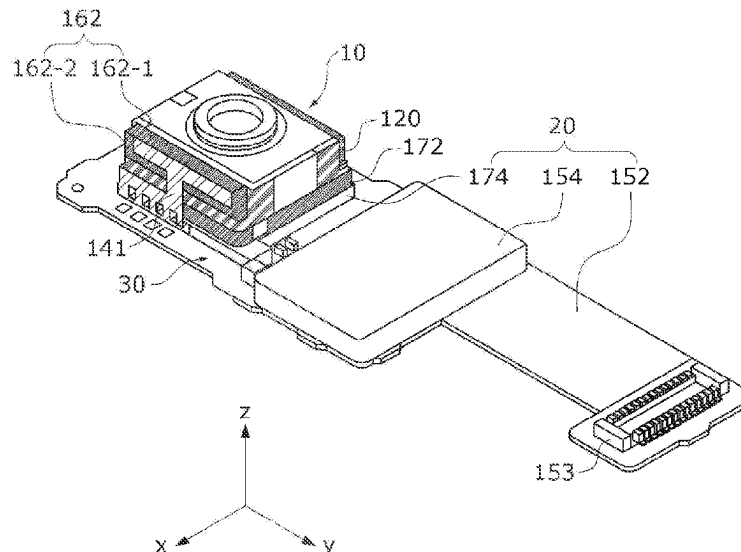
(57)

ABSTRACT

Disclosed is a camera module comprising: a lens assembly comprising a liquid lens which comprises a first liquid and a second liquid forming the surface; a gyro sensor for outputting inclination information; and a controller for adjusting the surface by applying a driving signal to the liquid lens on the basis of focal information and the inclination information, wherein the controller applies different weights to the driving signal in accordance with the angular velocity of the liquid lens obtained on the basis of the inclination information.

13 Claims, 26 Drawing Sheets

100



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G02B 27/64 (2006.01)
G03B 13/36 (2021.01)
H04N 23/52 (2023.01)
H04N 23/67 (2023.01)
H04N 23/68 (2023.01)
G02B 7/02 (2021.01)

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Fig. 1

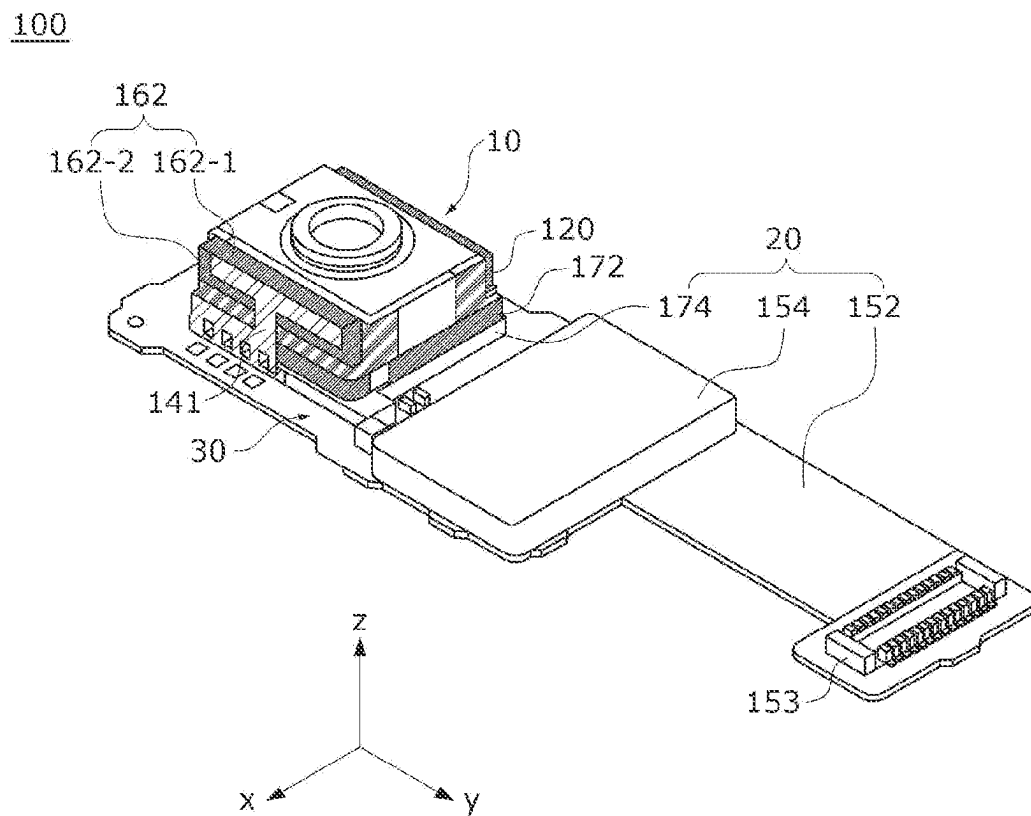


Fig. 2

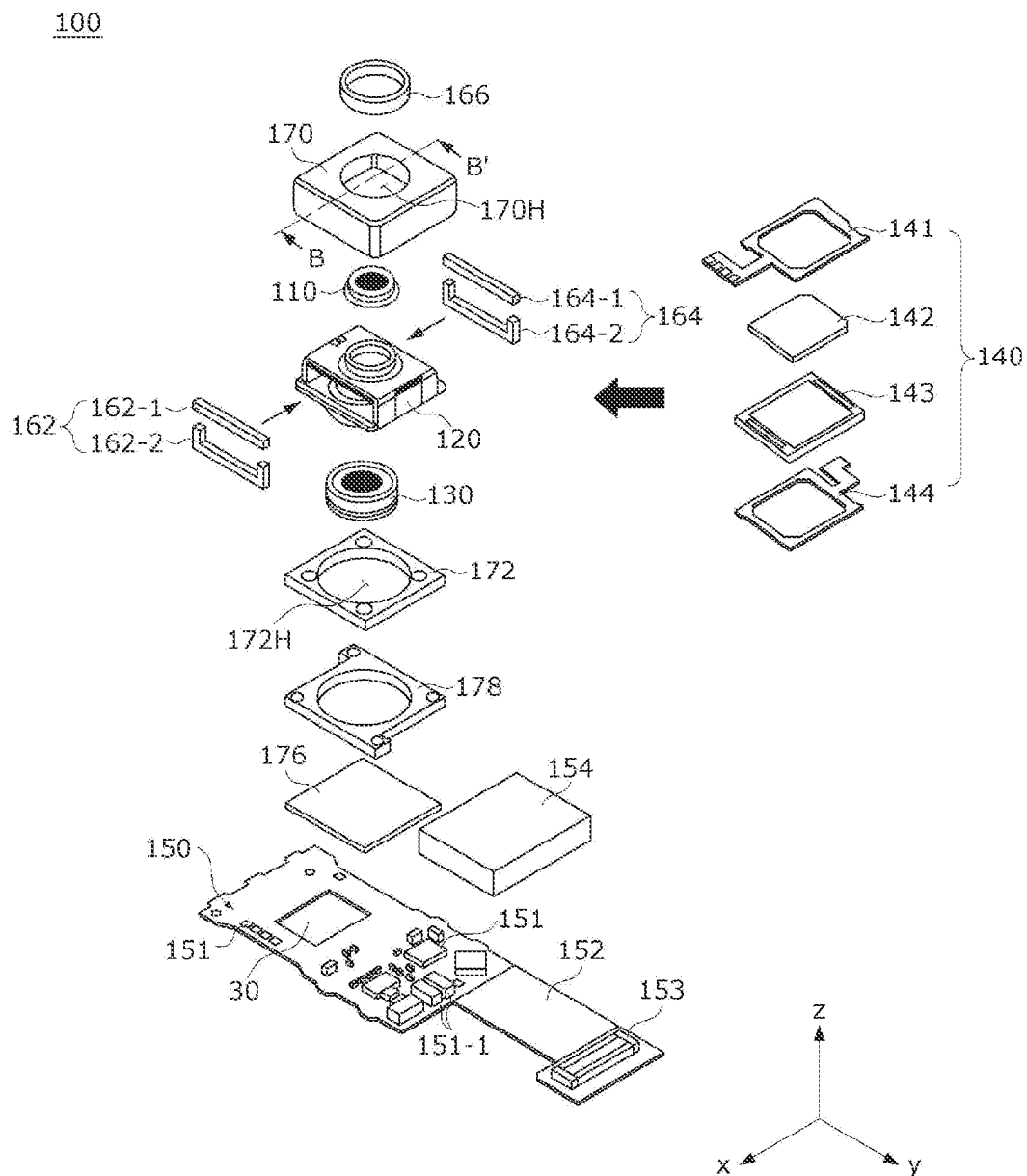


Fig. 3

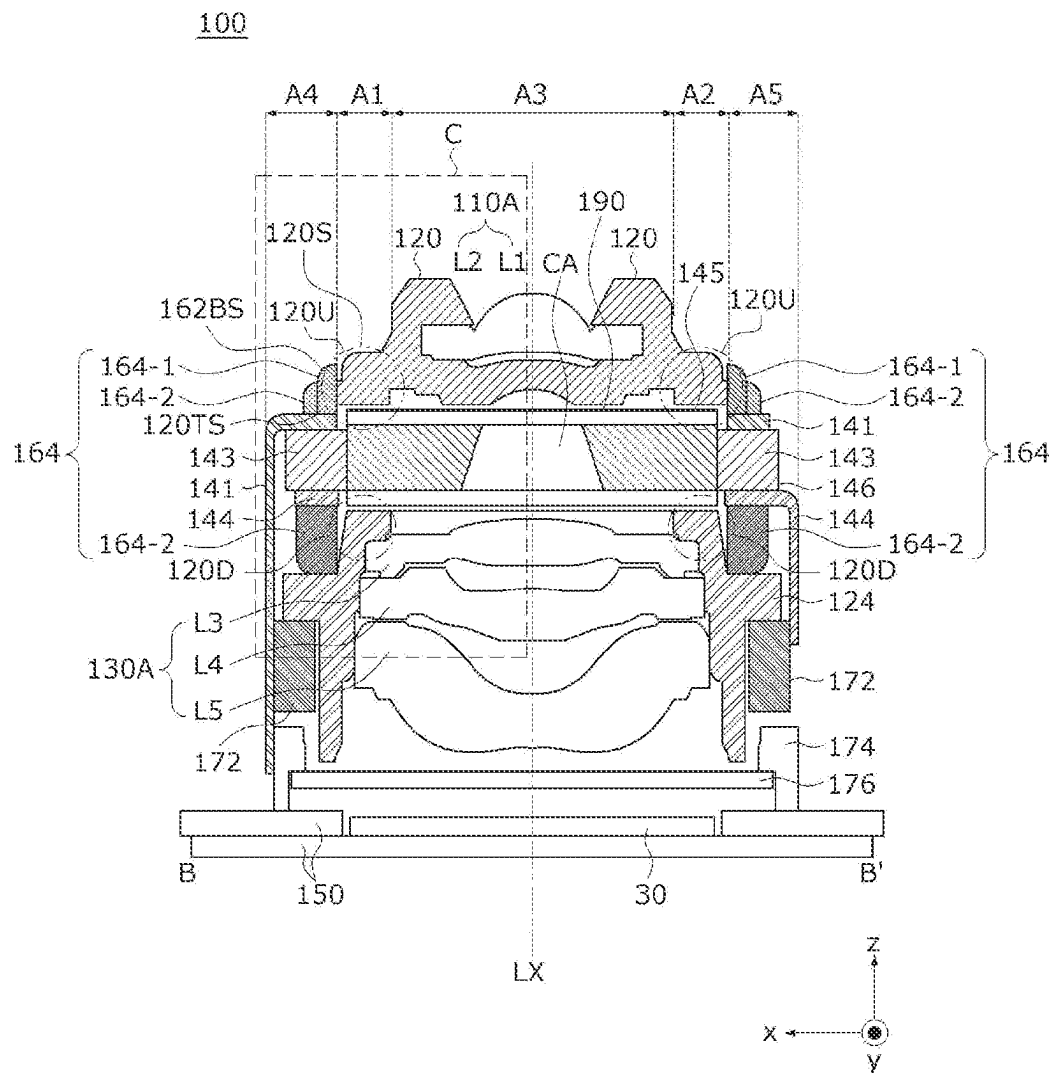


Fig. 4

140

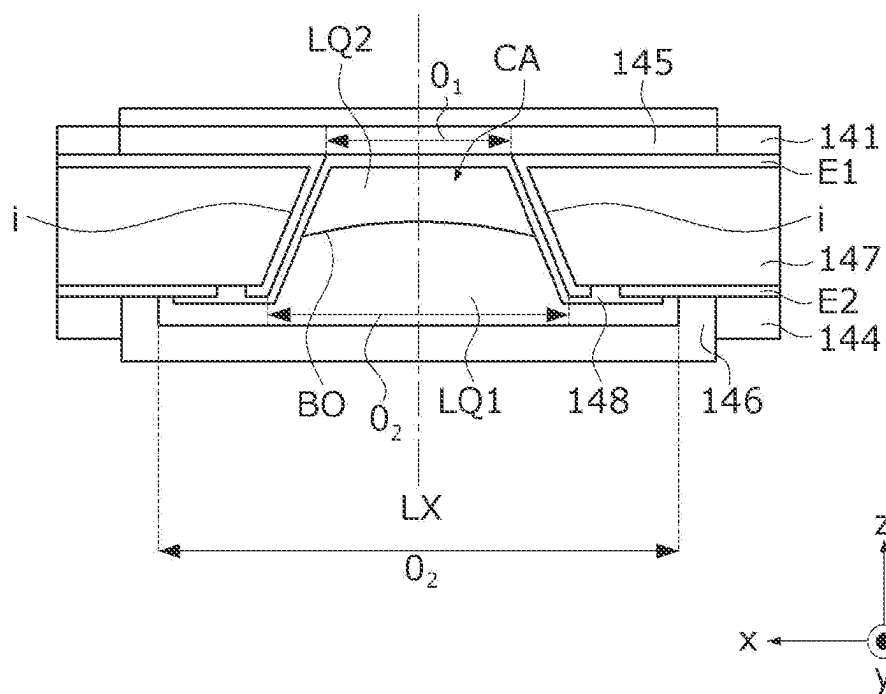


Fig. 5a

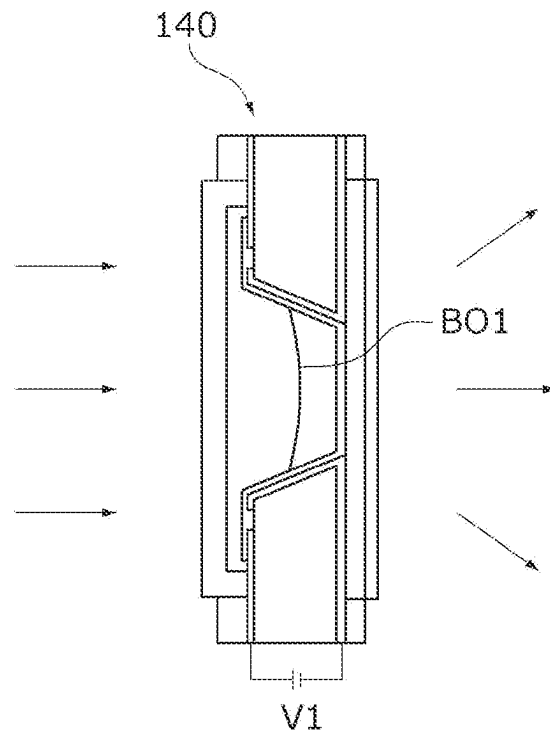


Fig. 5b

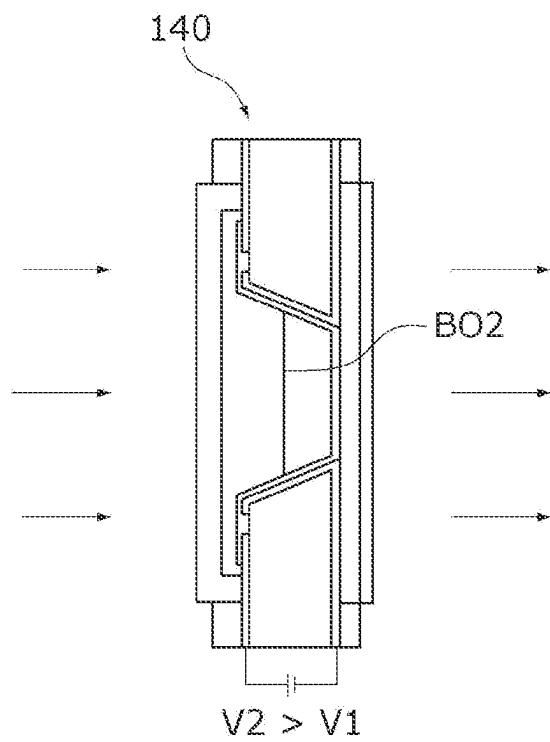


Fig. 5c

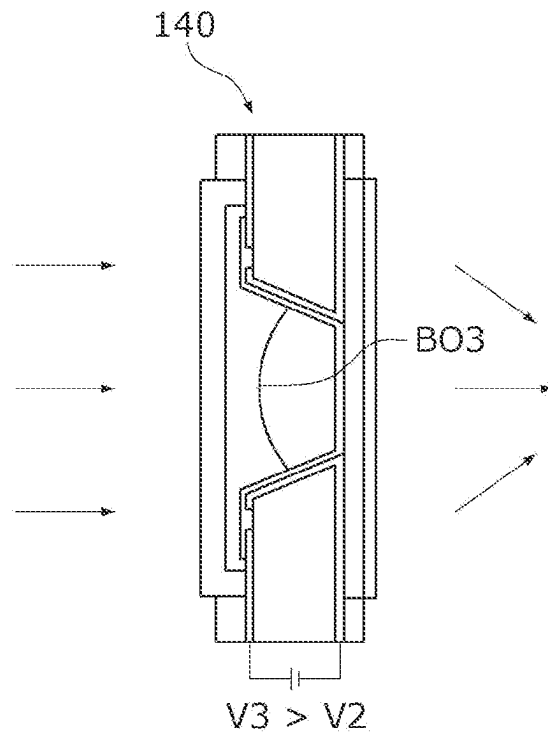


Fig. 5d

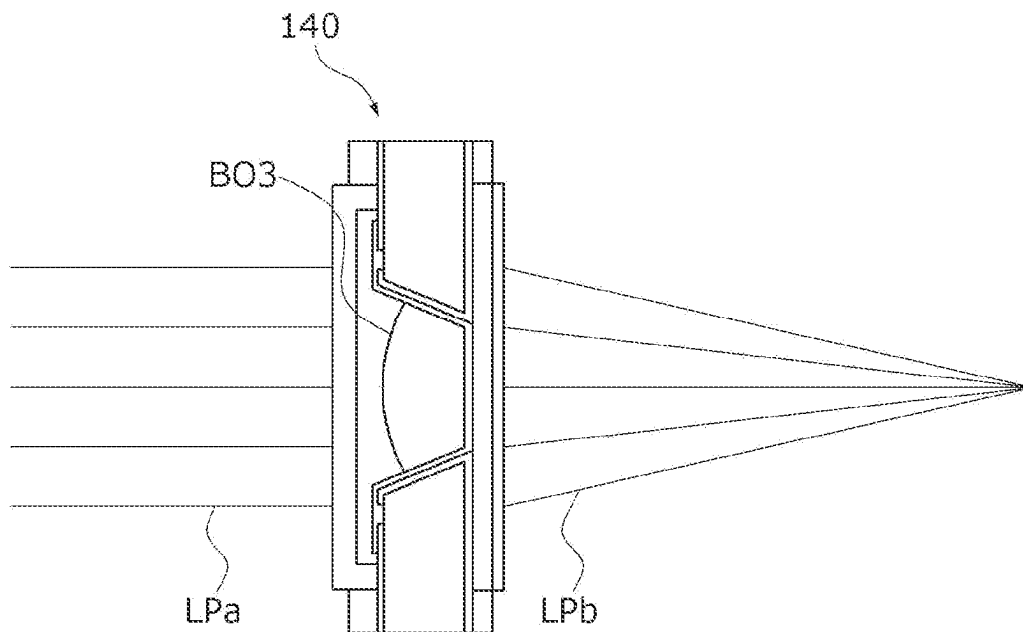


Fig. 5e

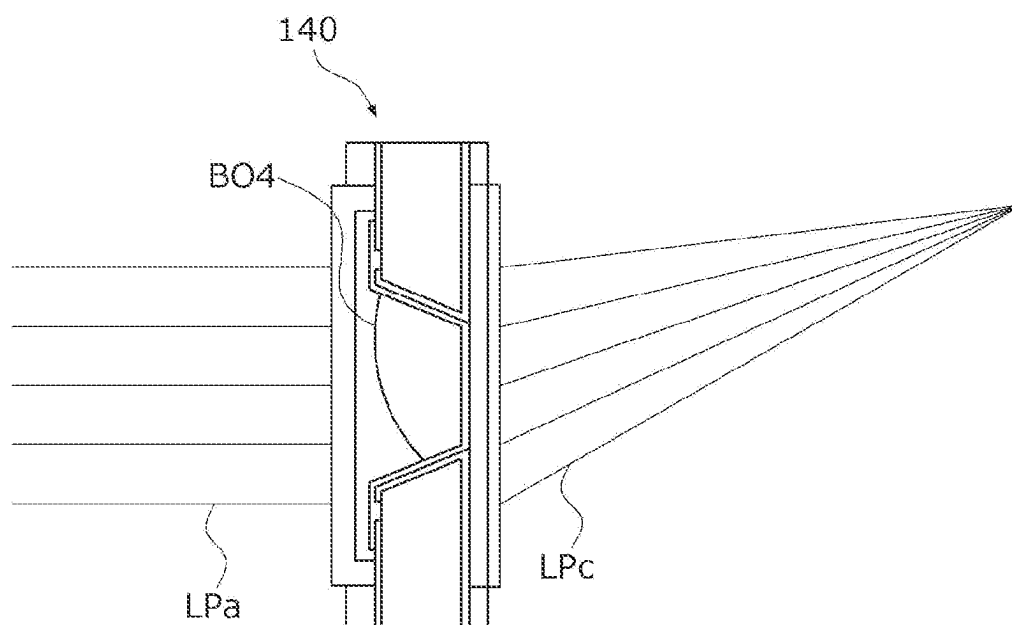


Fig. 6

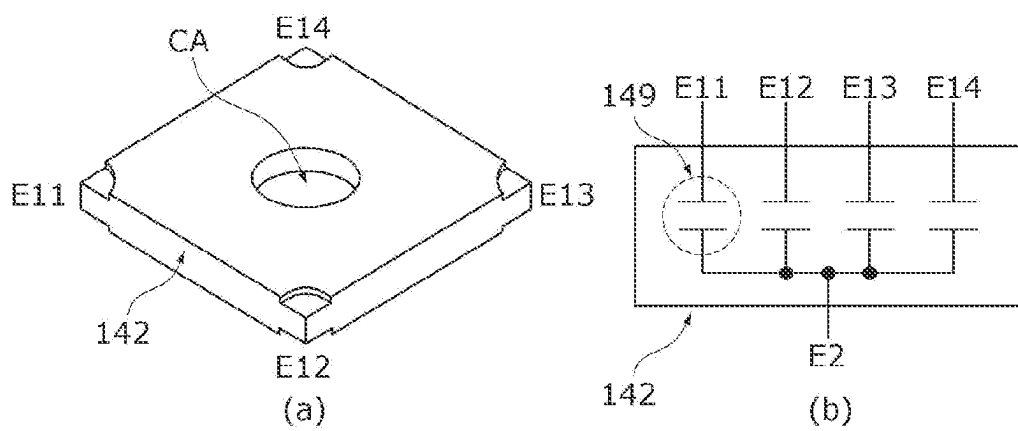


Fig. 7a

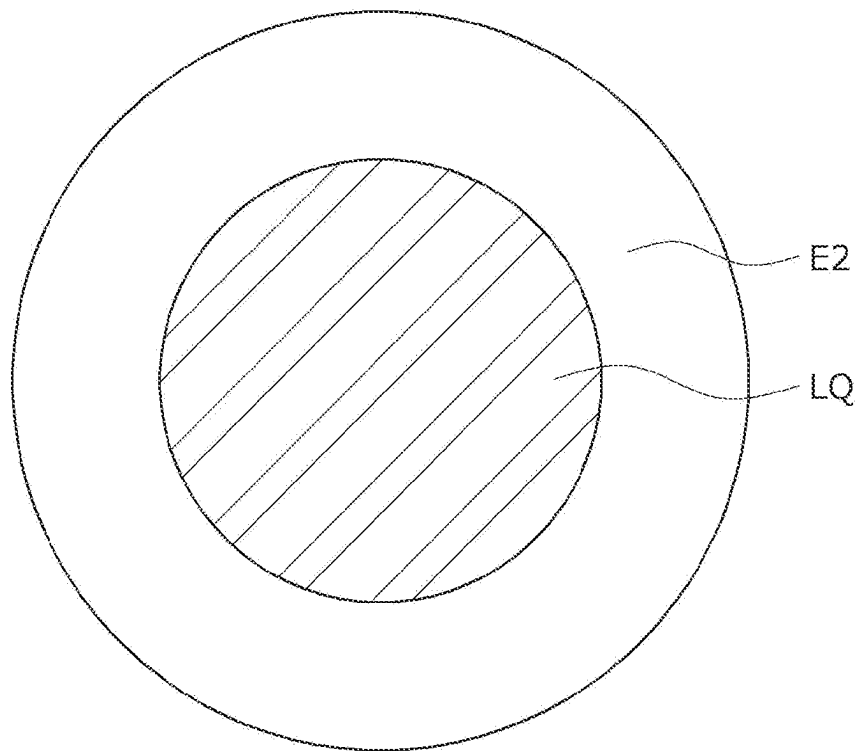


Fig. 7b

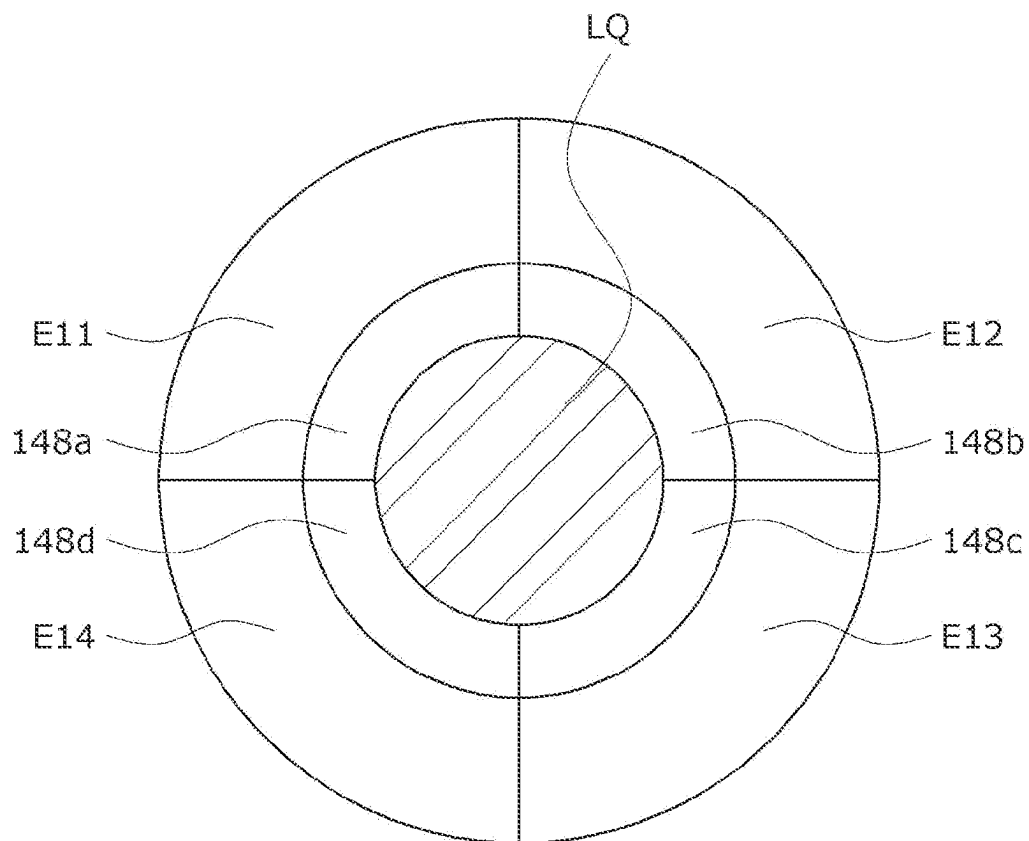


Fig. 7c

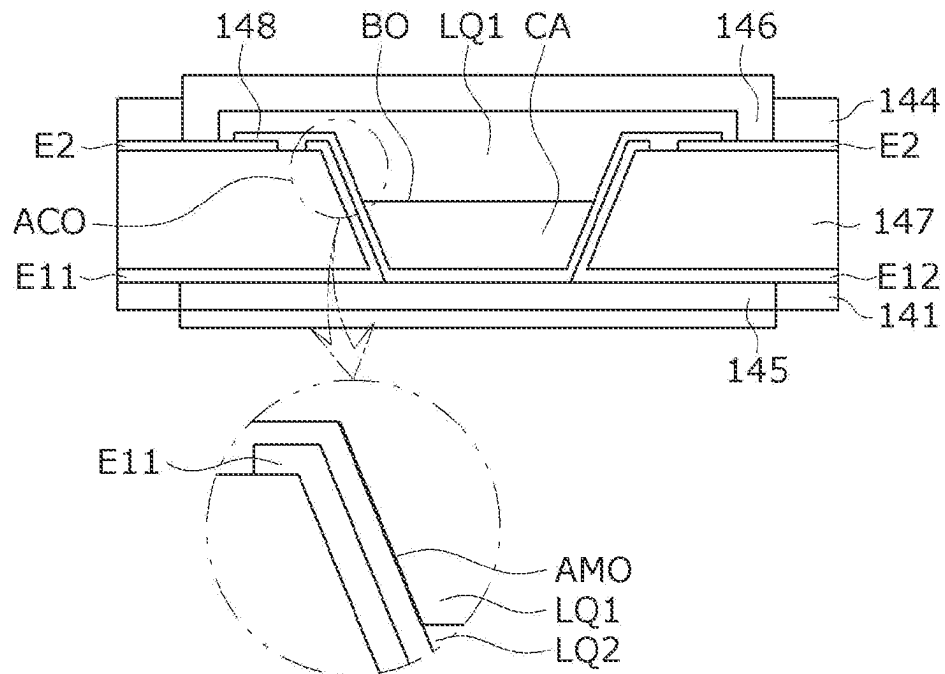


Fig. 8a

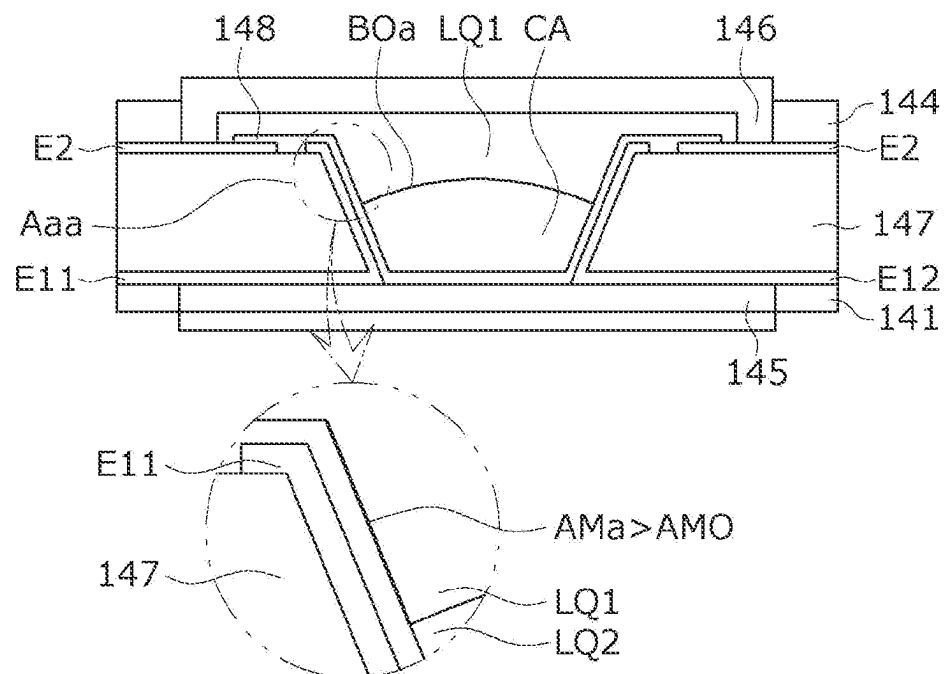


Fig. 8b

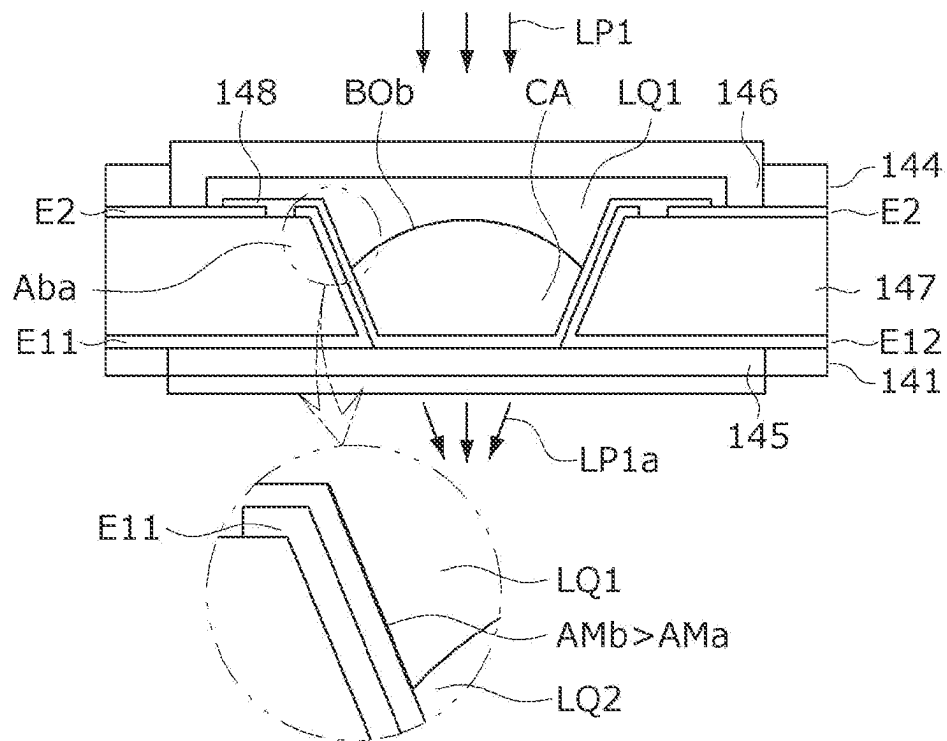


Fig. 8c

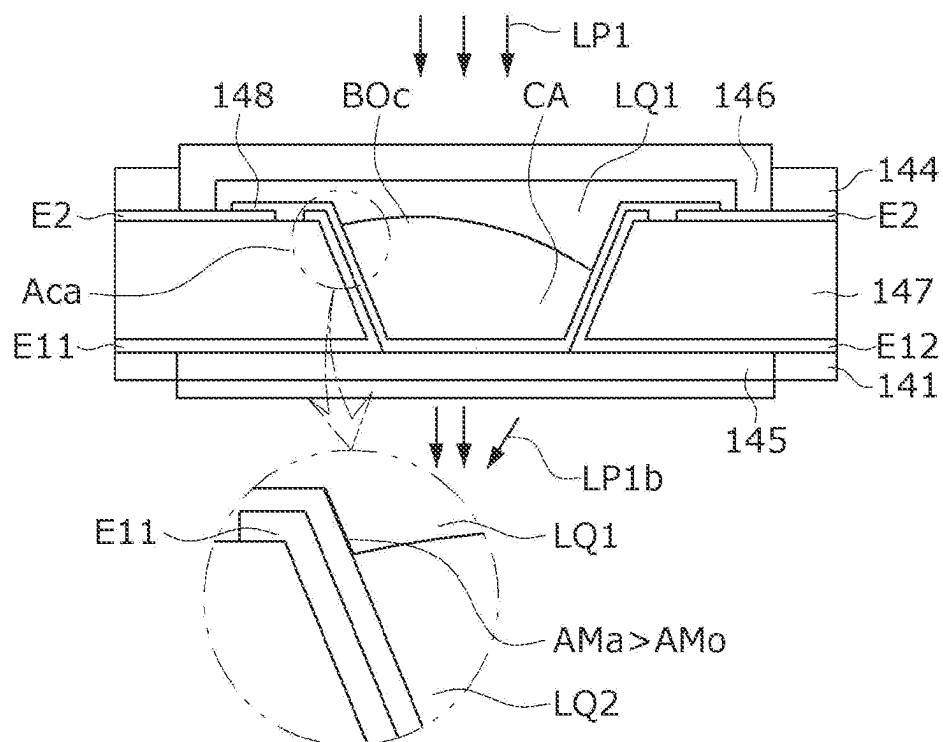


Fig. 8d

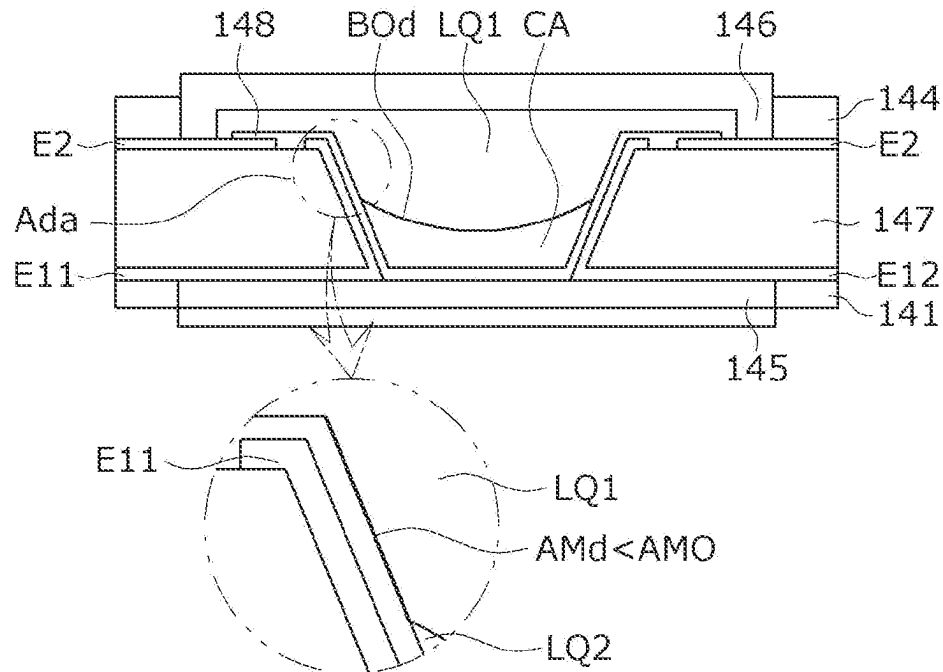


Fig. 8e

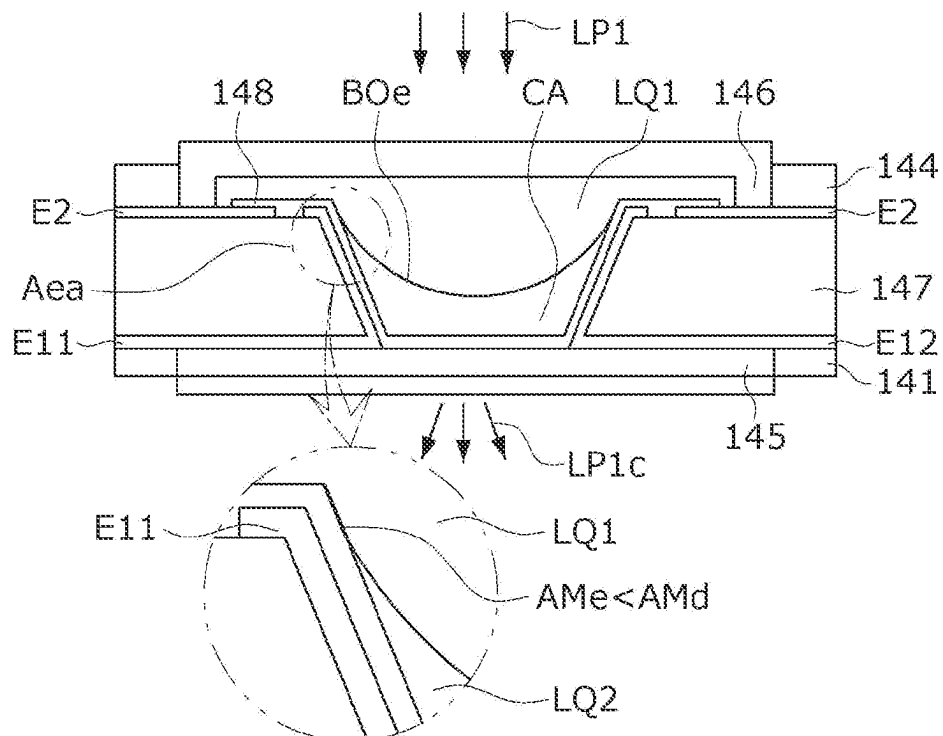


Fig. 9

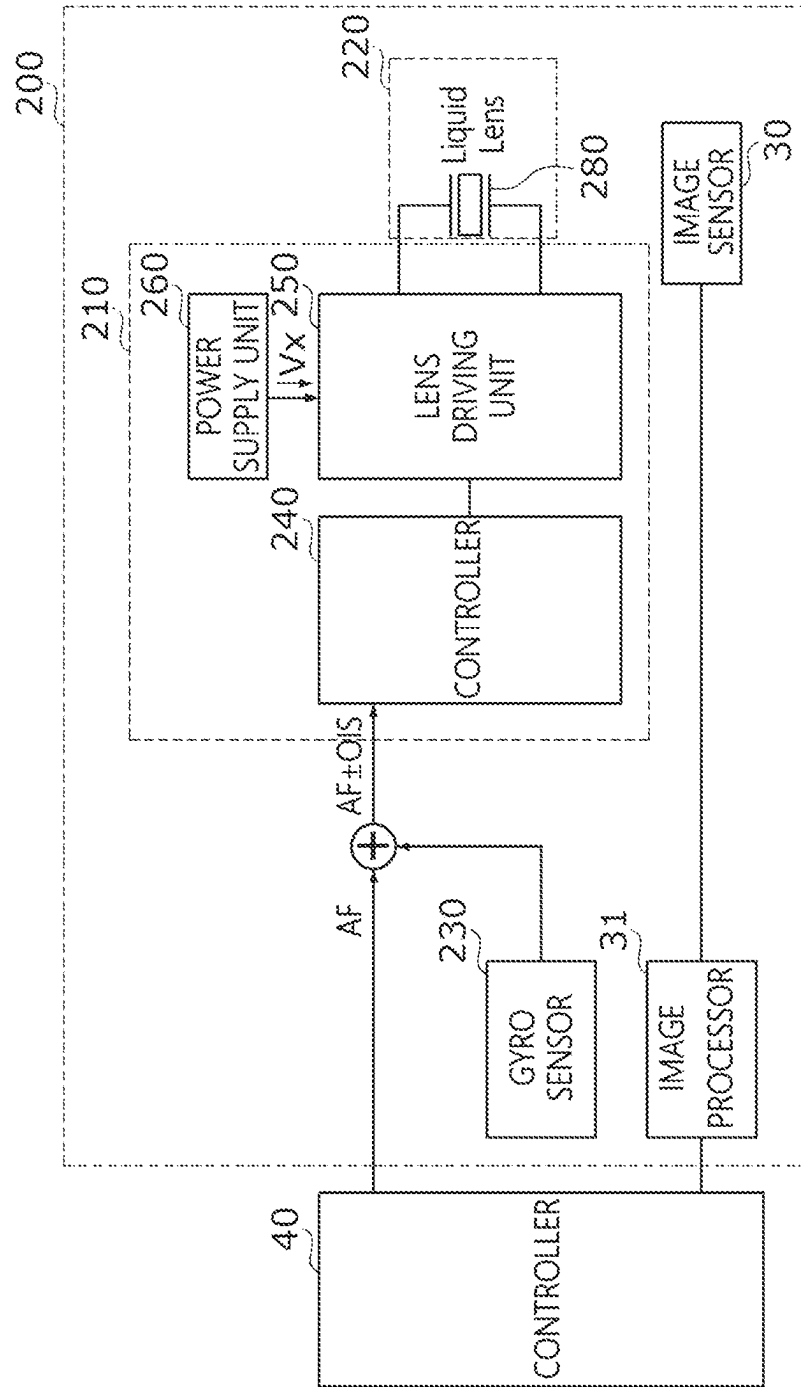


Fig. 10

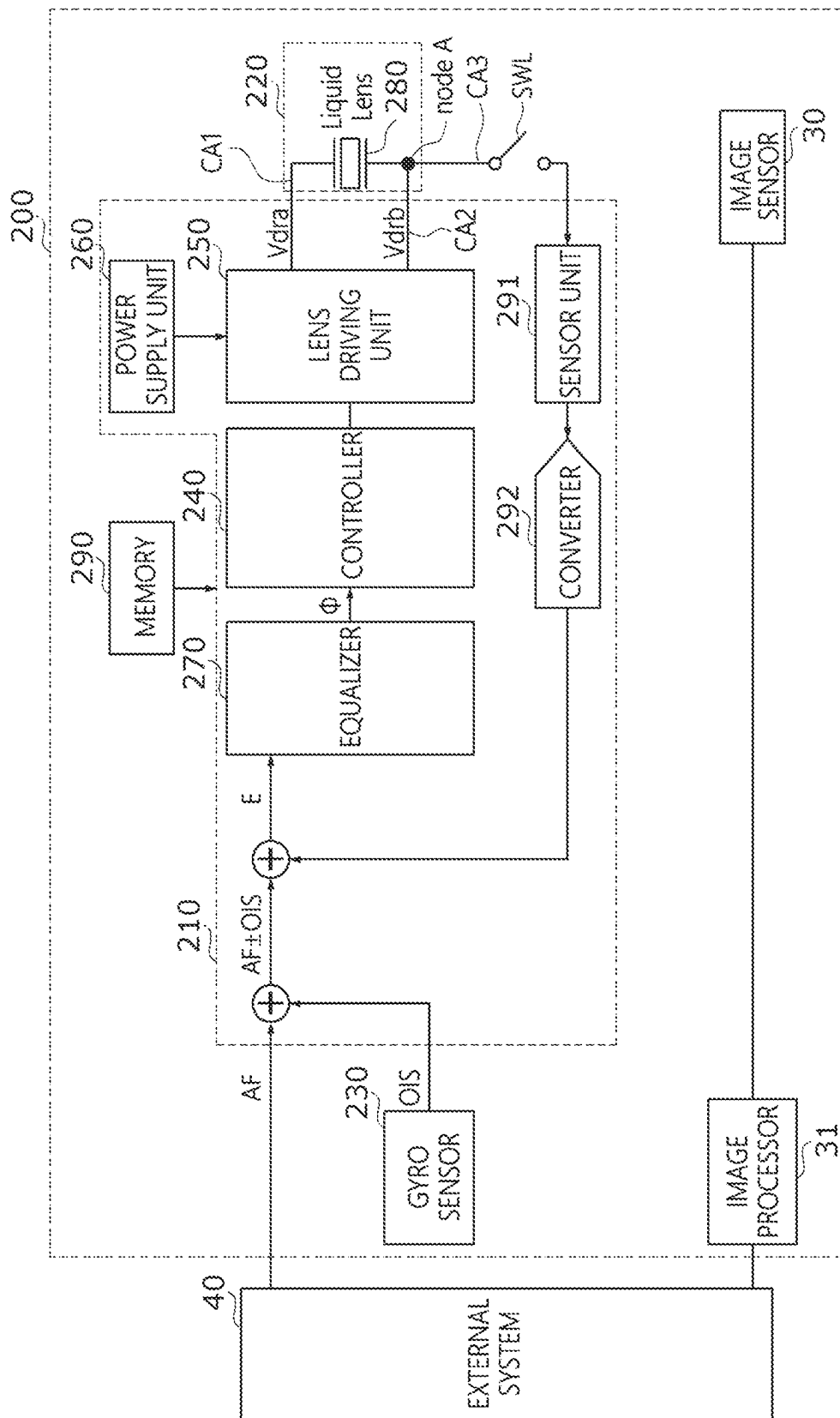
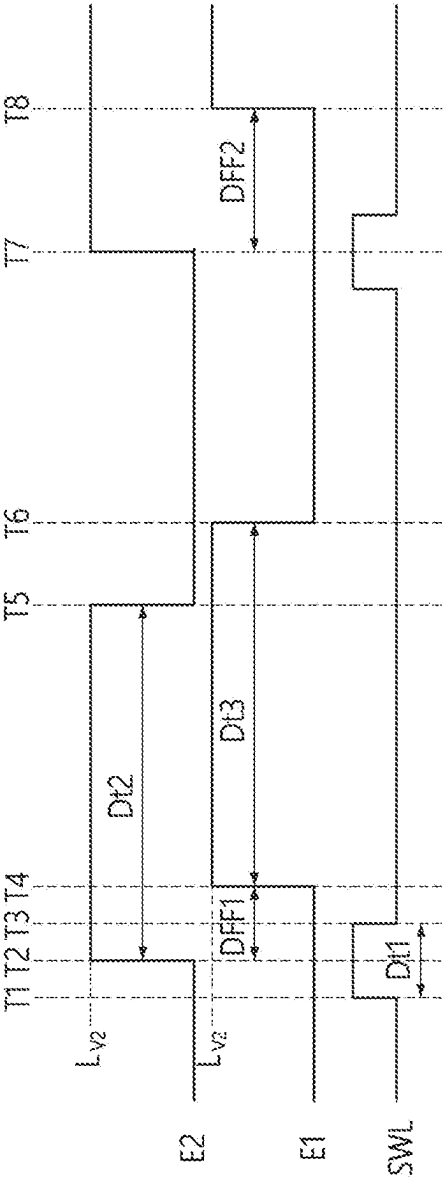


Fig. 11



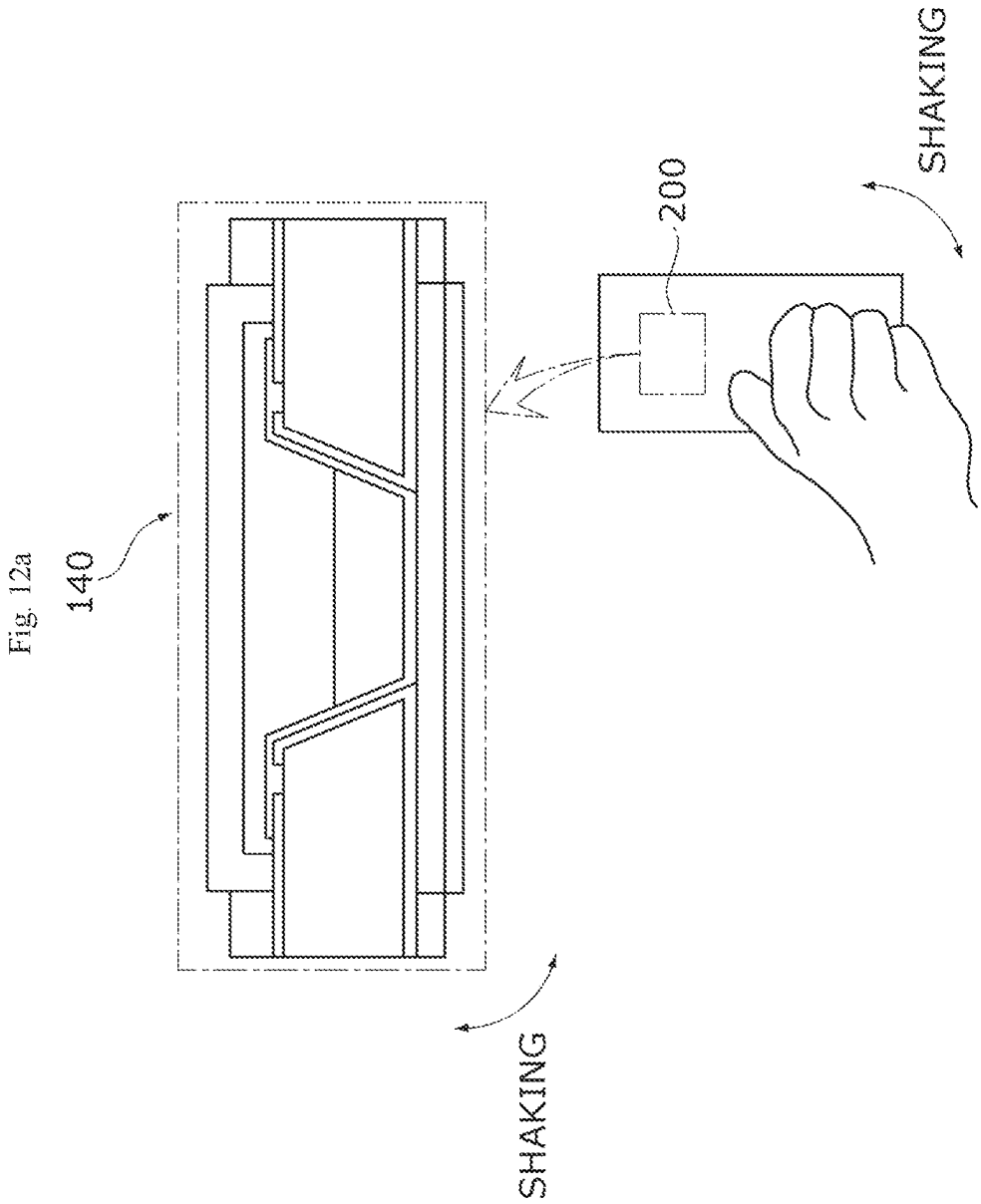


Fig. 12b

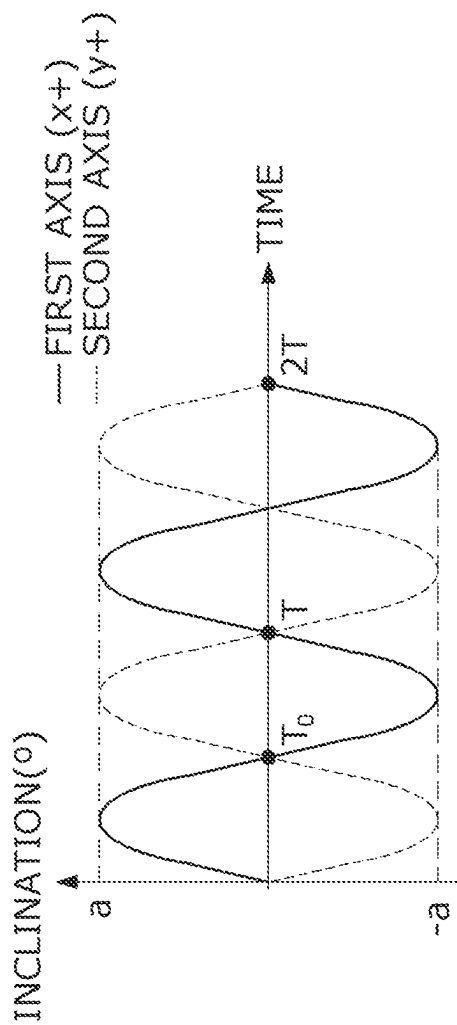


Fig. 13a

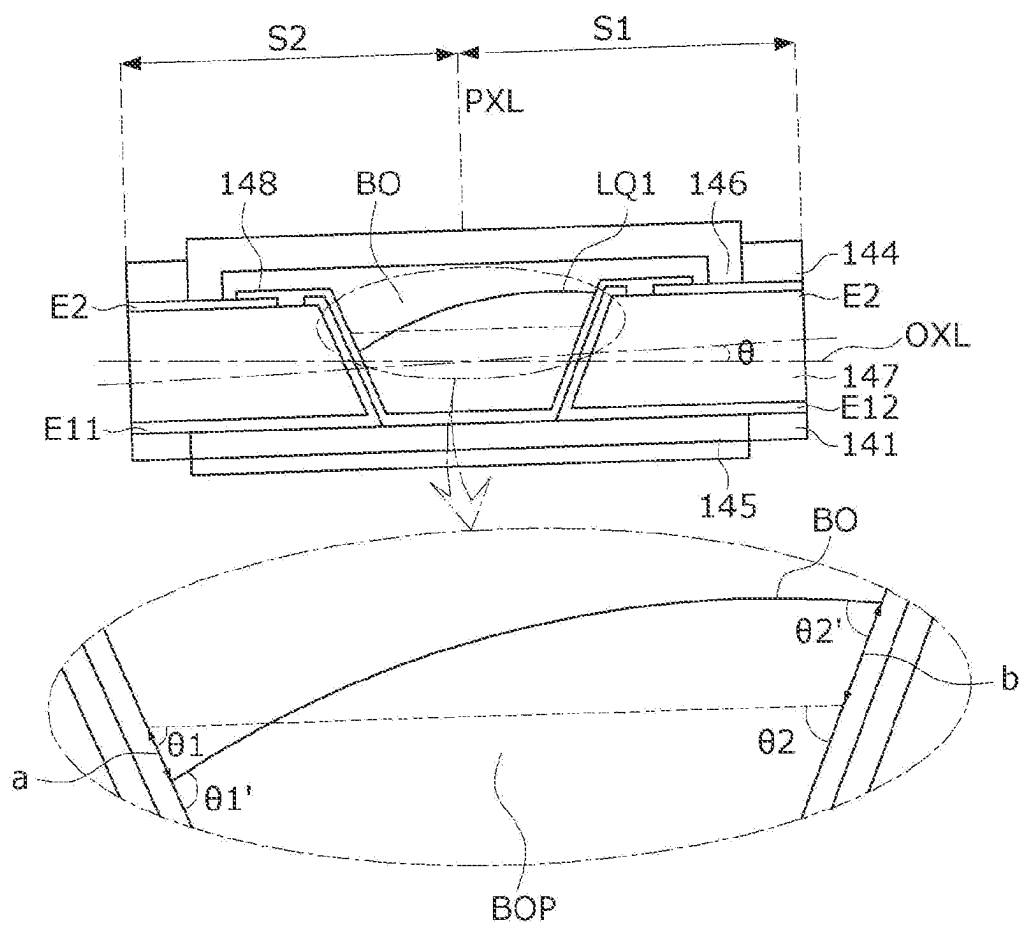


Fig. 13b

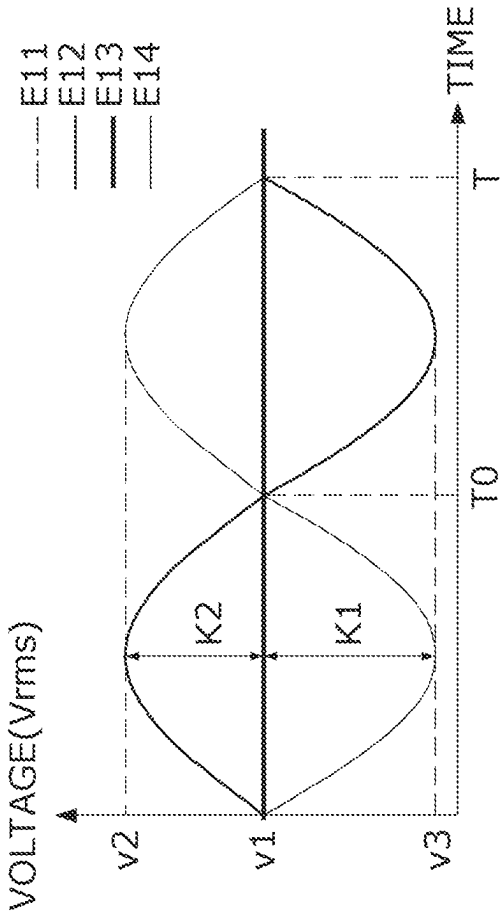


Fig. 13c

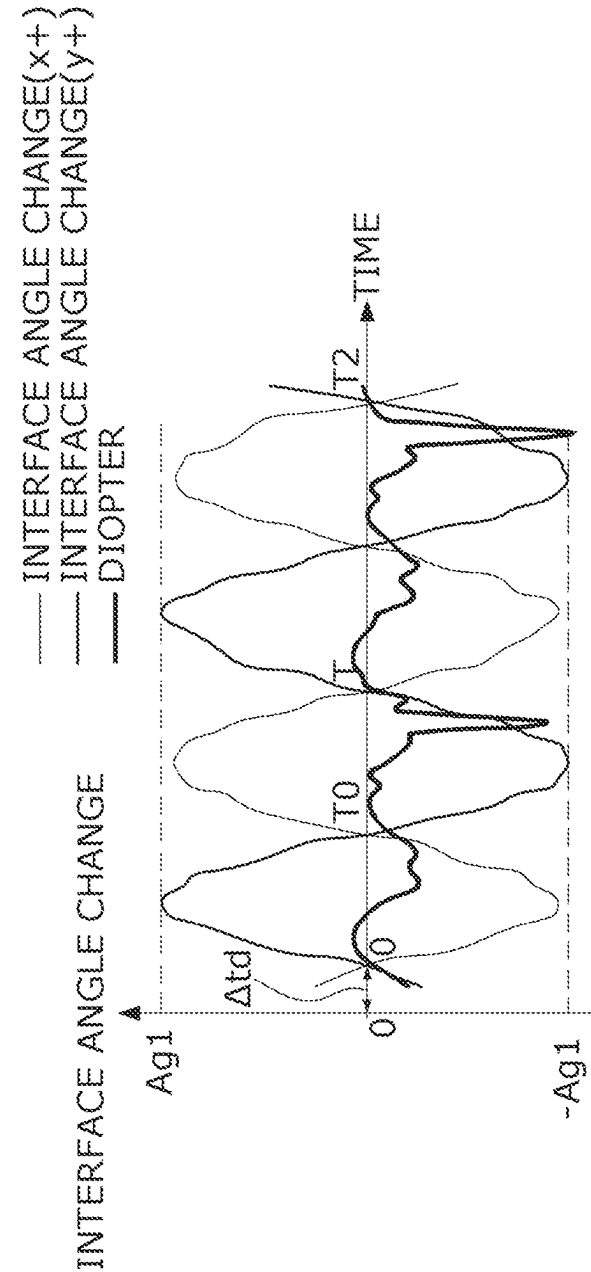


Fig. 14

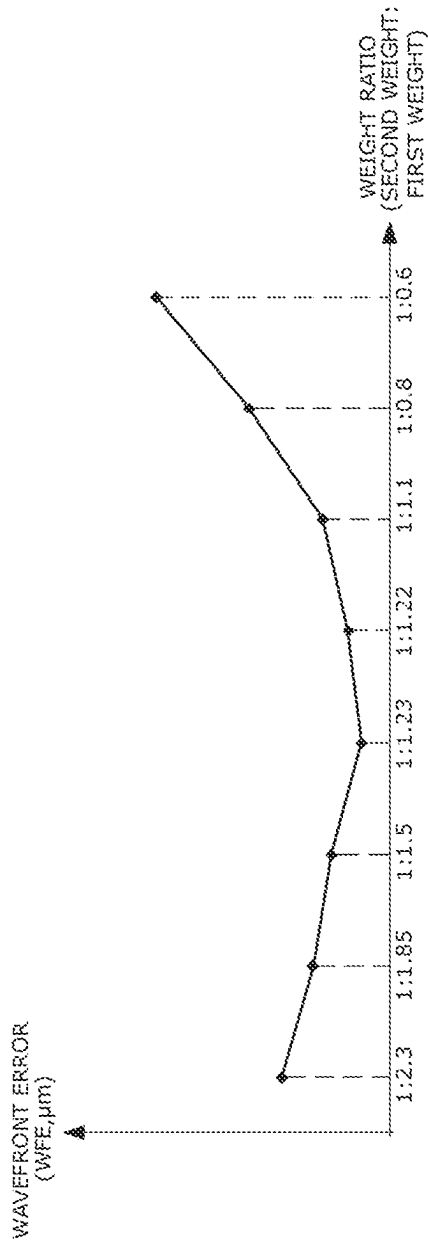


Fig. 15

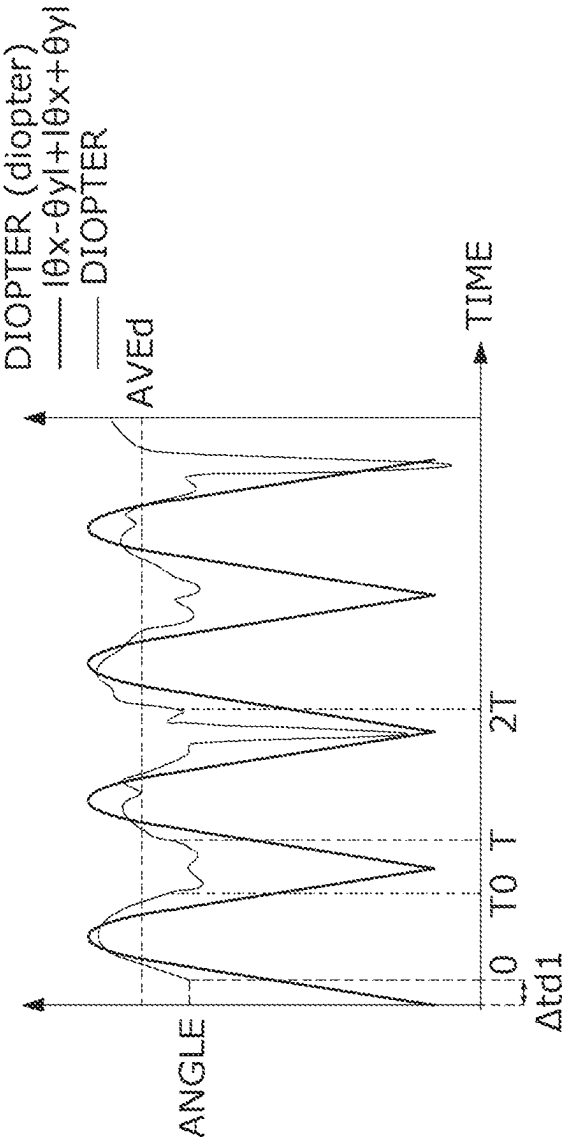


Fig. 16

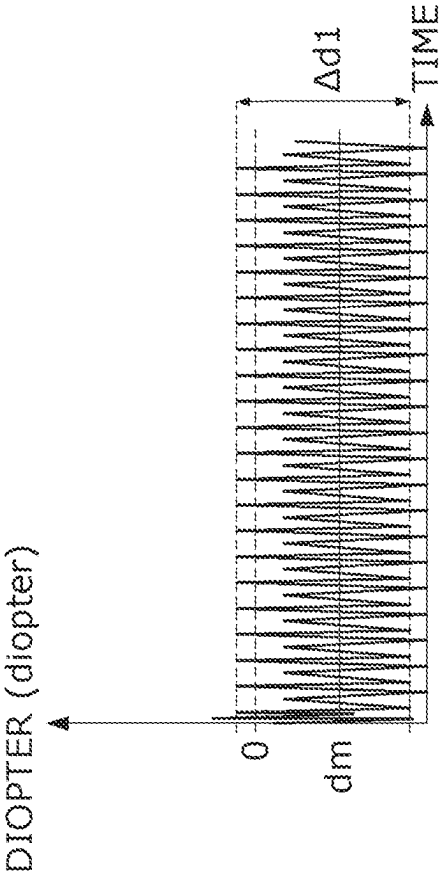


Fig. 18

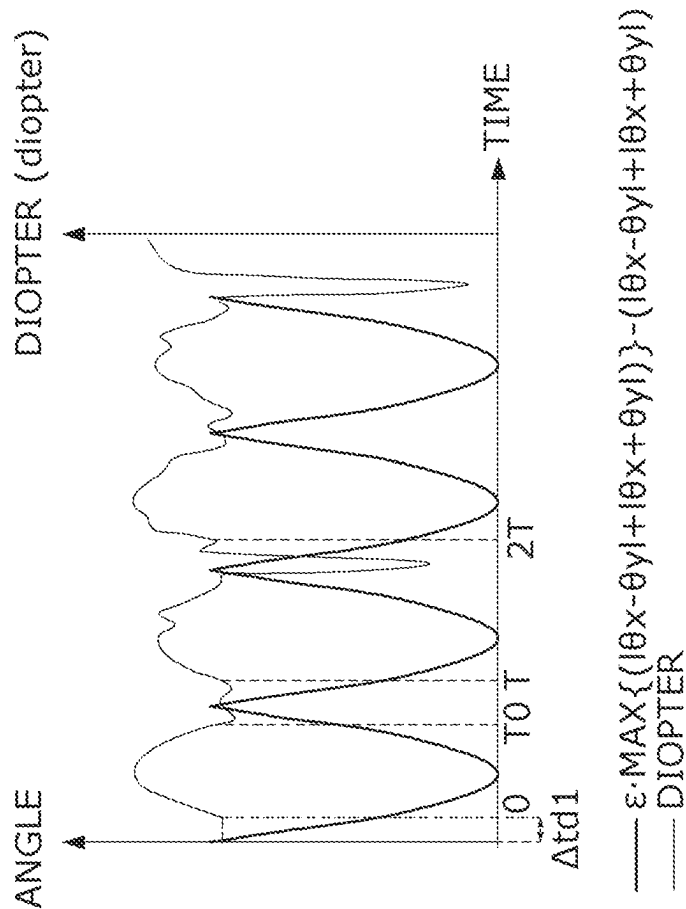
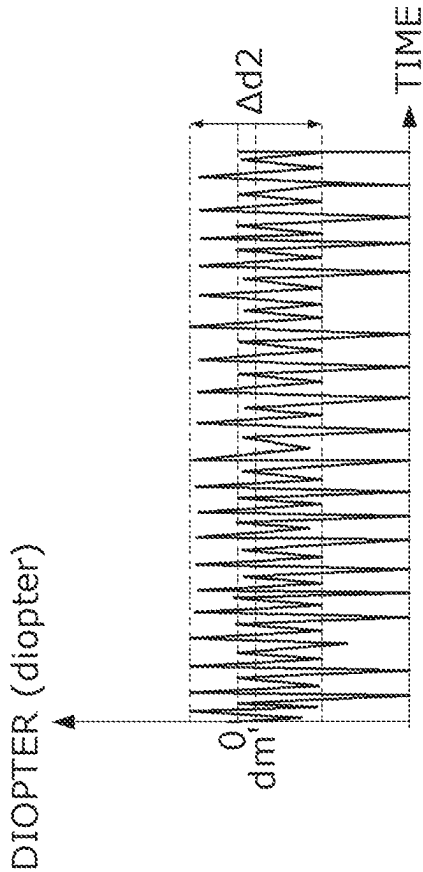


Fig. 19



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CAMERA MODULE

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is the U.S. national stage application of International Patent Application No. PCT/KR2020/017105, filed Nov. 27, 2020, which claims the benefit under 35 U.S.C. § 119 of Korean Application No. 10-2019-0155782, filed Nov. 28, 2019, the disclosures of each of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

An embodiment relates to a camera module.

BACKGROUND ART

Users of portable devices demand optical devices that have high resolution, are small in size, and have various photographing functions. For example, the various photographing functions may include at least one of an optical zoom-in/zoom-out function, an auto-focusing (AF) function, and a hand-tremor compensation or optical image stabilizer (OIS) function.

Conventionally, in order to realize the above-described various photographing functions, a method of combining a plurality of lenses and directly moving the combined lenses is used. However, in the case in which the number of lenses is increased as described above, the size of the optical device may be increased.

The auto-focusing function and the hand-tremor compensation function are performed by moving or tilting several lenses, which are fixed to a lens holder and are aligned with an optical axis, along the optical axis or in a direction perpendicular to the optical axis, and to this end, a separate lens driving device is required to drive a lens assembly composed of a plurality of lenses. However, the lens driving device has high power consumption, and in order to protect the lens driving device, there is a problem in that the overall size of the conventional camera module is increased because a cover glass needs to be separately added to the camera module. In order to solve this problem, research has been conducted on a liquid lens unit, which performs the auto-focusing function and the hand-tremor compensation function by electrically adjusting a curvature of an interface of two types of liquids.

However, the interface of the liquid lens is variously deformed according to the temperature, and thus improvements in accuracy are required.

DISCLOSURE

Technical Problem

An embodiment is directed to providing a camera module that includes a lens capable of adjusting a position of an interface positioned between two liquids according to electrical energy and is capable of compensating for a change of a diopter caused by deformation of the interface generated according to a temperature of a structure in which the two liquids included in the lens may be housed.

An embodiment is also directed to providing a camera module capable of adjusting a temperature to correspond to a change in a spatial frequency response (SFR) according to a temperature change of a liquid lens, and adjusting a driving voltage that may adjust the movement of an interface of

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liquids according to the temperature in order to overcome the characteristics and tendency of the liquid lens, in which a degree of curvature and bias of the interface formed by the two liquids included in the lens are adjusted according to a supply voltage and the resolution of the lens is lowered due to a thermal expansion coefficient according to the temperature change.

Objectives to be solved by the embodiment are not limited to the above-described objective and will include objectives or effects which may be identified by solutions for the objectives or the embodiments described below.

Technical Solution

A camera module according to an embodiment includes a lens assembly including a liquid lens having a first liquid and a second liquid that form an interface, a gyro sensor configured to output inclination information, and a controller configured to adjust the interface by applying a driving signal to the liquid lens on the basis of focus information and the inclination information, wherein the controller applies different weights to the driving signal according to an angular velocity of the liquid lens obtained on the basis of the inclination information. The controller may apply a first weight to the driving signal when the angular velocity of the liquid lens increases, and apply a second weight to the driving signal when the angular velocity of the liquid lens decreases, wherein the first weight may be greater than the second weight.

The controller may vary a time difference between a plurality of pulses applied to the liquid lens according to the first weight and the second weight.

The time difference may include a first delay corresponding to the first weight and a second delay corresponding to the second weight, wherein the first delay may be greater than the second delay.

A ratio of the second weight and the first weight may be in a range of 1:1.1 to 1:1.5.

The controller may compensate for a difference between the first weight and the second weight in response to a half cycle of the driving signal.

The controller may increase an average voltage difference of the driving signal for each cycle during compensation.

The controller may be driven according to Equation 1 below,

$$\begin{bmatrix} V_x^+ \\ V_x^- \\ V_y^+ \\ V_y^- \end{bmatrix} = V_0 \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} +G_{til}(g1 \text{ or } g2) & 0 & \varepsilon \\ -G_{til}(g1 \text{ or } g2) & 0 & \varepsilon \\ 0 & +G_{til}(g1 \text{ or } g2) & \varepsilon \\ 0 & -G_{til}(g1 \text{ or } g2) & \varepsilon \end{bmatrix} \quad \text{Equation 1}$$

$$\left[\begin{array}{c} \theta_x \\ \theta_y \\ \max_{\frac{1}{2} \leq \frac{1}{F}} \left[|\theta_x - \theta_y| + |\theta_x + \theta_y| - (|\theta_x - \theta_y| + |\theta_x + \theta_y|) \right] \end{array} \right]$$

(where V_x^+ , V_x^- , V_y^+ , and V_y^- are each a voltage of the driving signal applied to each individual electrode, V_0 is a voltage applied to the individual electrode to set a focus according to auto-focusing (AF) or a voltage before the compensation for shaking, G_{til} is a linear correlation coefficient between an optical inclination of the liquid lens and a voltage offset between opposite electrodes (e.g., X^+ and X^- or Y^+ and Y^-), $g1$ is the second weight, $g2$ is the first weight, and F is a focus compensation parameter according to the first and

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second weights in the controller. In addition, θ_x is an x-axis angle, and θ_y is a y-axis angle.)

The camera module may further include a sensor unit configured to detect interface information corresponding to a curvature of the interface.

The sensor unit may detect a capacitance corresponding to a size or a change in an area of the interface.

The camera module may further include an image sensor configured to receive light passing through the lens assembly, and the focus information may be based on image information output from the image sensor.

The controller may change the weights equally after a predetermined time.

The liquid lens may include a plate including a cavity in which a conductive liquid and a non-conductive liquid are disposed while forming an interface, a plurality of individual terminals disposed below the plate, and a common terminal disposed above the plate.

The driving signal may be applied to the plurality of individual terminals.

The driving signal may include a signal applied to at least one of the plurality of individual terminals and the common electrode for a predetermined cycle.

Advantageous Effects

An embodiment provides a camera module capable of inhibiting a decrease in resolution or accuracy caused by a change in an interface of a liquid lens according to a temperature.

Various advantages and effects of the present invention are not limited to the above description and can be more easily understood through the description of specific exemplary embodiments of the present invention.

DESCRIPTION OF DRAWINGS

FIG. 1 is a schematic perspective view of a camera module according to an embodiment.

FIG. 2 is an exploded perspective view of the camera module according to the embodiment.

FIG. 3 is a cross-sectional view of the camera module according to the embodiment.

FIG. 4 is a cross-sectional view of a liquid lens unit according to the embodiment.

FIGS. 5A to 5E are views for describing a method of driving the liquid lens unit.

FIG. 6 is a view for describing the liquid lens unit in which an interface is adjusted according to a driving voltage.

FIGS. 7A to 7C are views illustrating a structure of the liquid lens unit.

FIGS. 8A to 8E are views for describing a variation in a lens curvature of the liquid lens unit.

FIG. 9 is a block diagram of the camera module according to the embodiment.

FIG. 10 illustrates a modified example of FIG. 9.

FIG. 11 is a view referenced in the description of FIGS. 9 and 10.

FIGS. 12A and 12B are views for describing a driving of the camera module in response to shaking.

FIGS. 13A and 13B are views for describing a case in which different driving signals are applied according to weights.

FIG. 13C is a view illustrating a change in an interface according to the application of driving signals having different weights.

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FIG. 14 is a view illustrating a wavefront error (WFE) according to weights.

FIG. 15 is a view illustrating a change in diopter for a half cycle of the driving signal.

FIG. 16 is a view for describing diopter defocusing.

FIG. 17 is a view illustrating a variation of the driving signal during compensation.

FIG. 18 is a view for describing compensation according to a weight difference.

FIG. 19 is a view illustrating an average voltage value according to compensation.

MODES OF THE INVENTION

Hereinafter, exemplary embodiments of the present invention will be described in detail with reference to the accompanying drawings.

However, the technical spirit of the present invention is not limited to some embodiments to be disclosed below but may be realized in many different forms, and one or more components of each of the embodiments may be selectively combined and substituted within the scope of the present invention.

In addition, unless clearly and expressly defined herein, the terms (including technical and scientific terms) used in the embodiments of the present invention have the same meaning as commonly understood by one of ordinary skill in the art. It should be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the related art.

Further, the terms used in the embodiments of the present invention are provided only to describe embodiments of the present invention and not for purposes of limitation.

In the present specification, the singular forms include the plural forms unless the context clearly indicates otherwise, and the phrase "at least one element (or one or more elements) of an element A, an element B, and an element C," should be understood as including the meaning of at least one of all possible combinations of the element A, the element B, and the element C.

Further, in describing components of the embodiments of the present invention, the terms such as first, second, A, B, (a), (b), and the like may be used.

These terms are merely for distinguishing one component from another component, and the property, order, sequence, and the like of the corresponding components are not limited by the terms.

In addition, it will be understood that when one component is referred to as being "connected" or "coupled" to another component, the component may not only be directly connected or coupled to another component but may also be connected or coupled to another component with still another component presented between one component and another component.

Further, when one component is referred to as being formed or disposed "on (above)" or "under (below)" another component, the terms "on (above)" or "under (below)" includes both of a case in which the two components are in direct contact with each other or a case in which one or more components are (indirectly) formed or disposed between the two components. In addition, the term "on (above)" or "under (below)" includes a case in which another component is disposed in an upward direction or a downward direction with respect to one component.

The term "auto-focusing function" used below is defined as a function of automatically focusing on an object by

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moving a lens in an optical-axis direction to adjust a distance to an image sensor according to a distance to the object, in order to allow the image sensor to acquire a clear image of the object. Meanwhile, the “auto-focusing” may correspond to “auto focus (AF).”

The term “hand-tremor compensation function” used below is defined as a function of moving a lens and/or an image sensor to offset vibrations (movement) generated in the image sensor by an external force. Meanwhile, the “hand-tremor compensation” may correspond to “optical image stabilization (OIS).”

FIG. 1 is a schematic perspective view of a camera module according to an embodiment.

Referring to FIG. 1, a camera module 100 may include a lens assembly 10, a control circuit 20, and an image sensor 30.

First, the lens assembly 10 may include a plurality of lens units and a holder in which the plurality of lens units are accommodated. As described below, the plurality of lens units may include a liquid lens, and may further include a first lens unit or a second lens unit. The plurality of lens units may include the first and second lens units and a liquid lens unit.

The control circuit 20 serves to supply a driving voltage (or an operating voltage) to the liquid lens unit.

The control circuit 20 and the image sensor 30, which are described above, may be disposed on a single printed circuit board (PCB), but this is merely given by way of example, and the embodiment is not limited thereto.

When the camera module 100 according to the embodiment is applied to an optical device or optical instrument, the configuration of the control circuit 20 may be designed in different ways according to specifications required in the optical device. In particular, the control circuit 20 may be implemented as a single chip to reduce the magnitude of a driving voltage applied to the lens assembly 10. Thus, the size of an optical device mounted on a portable device can be further reduced. The detailed description thereof will be given below.

FIG. 2 is an exploded perspective view of the camera module according to the embodiment, FIG. 3 is a cross-sectional view of the camera module according to the embodiment, and FIG. 4 is a cross-sectional view of the liquid lens unit according to the embodiment.

Referring to FIGS. 2 and 3, the camera module 100 may include the lens assembly, a main substrate 150, and the image sensor 30. In addition, the camera module 100 may further include a first cover 170 and a middle base 172. In addition, the camera module 100 may further include one or more adhesive members 162 and 164 and a second cover 174. The one or more adhesive members serve to couple or fix a liquid lens unit 140 to a holder 120.

Here, the one or more adhesive members are exemplified as including all of a first adhesive member 162, a second adhesive member 164, and a third adhesive member 166, but the embodiment is not limited thereto. That is, according to another embodiment, the one or more adhesive members may also include only some of the first adhesive member 162, the second adhesive member 164, and the third adhesive member 166.

Further, according to the embodiment, at least one of components 110 to 190 of the camera module 100 illustrated in FIG. 2 may be omitted. Alternatively, at least one component different from the components 110 to 190 illustrated in FIG. 2 may be further added and included in the camera module 100.

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For convenience of description, the third adhesive member 166, the first cover 170, the second cover 174, and an optical layer 190 illustrated in FIG. 2 are omitted in FIG. 3.

Further, the lens assembly 10 (see FIG. 1) may include at least one of the liquid lens unit 140, the holder 120, the first lens unit 110, the second lens unit 130, the first adhesive member 162, the second adhesive member 164, and the third adhesive member 166. In addition, the lens assembly may be disposed above the main substrate 150.

In addition, in order to distinguish the first and second lens units from the liquid lens unit 140 in the lens assembly, the first lens unit 110 and the second lens unit 130 may be referred to as a “first solid lens unit” and a “second solid lens unit,” respectively.

The first lens unit 110 may be disposed on an upper side of the lens assembly and may be a region on which light is incident from the outside of the lens assembly. That is, the first lens unit 110 may be disposed above the liquid lens unit 140 in the holder 120. The first lens unit 110 may be implemented as a single lens and may be implemented as two or more lenses that are aligned along a central axis to form an optical system. Here, the central axis may be an optical axis LX of the optical system, which is formed by the first lens unit 110, the liquid lens unit 140, and the second lens unit 130 included in the camera module 100, and may be an axis parallel to the optical axis LX. The optical axis LX may be the same as an optical axis f of the image sensor 30.

That is, the first lens unit 110, the liquid lens unit 140, the second lens unit 130, and the image sensor 30 may be aligned and disposed along the optical axis LX through active alignment (AA). Here, the active alignment may be an operation of aligning the optical axes of the first lens unit 110, the second lens unit 130, and the liquid lens unit 140, and adjusting an axis relationship or a distance relationship between the image sensor 30 and the lens units 110, 130, and 140 in order to acquire an improved image.

In an embodiment, the active alignment may be performed by an operation of analyzing image data generated by receiving light, which is incident from a specific object, by the image sensor 30 through at least one of the first lens unit 110, the second lens unit 130, and the liquid lens unit 140. For example, the active alignment may be performed in the following sequence.

As an example, after active alignment (first alignment) that adjusts relative positions between the first and second lens units 110 and 130, which are fixed and mounted to the holder 120, and the image sensor 30 is completed, and active alignment (second alignment) that adjusts relative positions between the liquid lens unit 140, which is inserted into the holder 120, and the image sensor 30 may be performed. The first alignment may be performed while a gripper grips the middle base 172 and moves the middle base to various positions, and the second alignment may be performed while the gripper grips a spacer 143 of the liquid lens unit 140 and moves the spacer to various positions. However, the active alignment may be performed in a sequence different from the above-described sequence.

In addition, the holder 120 may include a holder upper region 120U disposed above the liquid lens unit 140 and a holder lower region 120D disposed below the liquid lens unit 140. In this case, the first and second adhesive members 162 and 164 may respectively couple the holder upper region 120U and the holder lower region 120D to the liquid lens unit 140.

In addition, when the first and second adhesive members **162** and **164** are disposed, the liquid lens unit **140** may be stably fixed and coupled to the holder **120**.

Further, the first lens unit **110** may include, for example, two lenses **L1** and **L2**, but this is exemplary, and the number of lenses included in the first lens unit **110** may be one or more.

In addition, an exposure lens may be disposed on an upper side of the first lens unit **110**. Here, the exposure lens may mean an outermost lens among the lenses included in the first lens unit **110**. That is, the lens **L1** positioned at the uppermost side of the first lens unit **110** may protrude upward and thus perform a function of the exposure lens. A surface of the exposure lens protrudes to the outside of the holder **120** and thus may be damaged. When the surface of the exposure lens is damaged, the quality of an image captured by the camera module **100** may be degraded. Accordingly, in order to inhibit or suppress damage to the surface of the exposure lens, a cover glass may be disposed, or a coating layer may be formed on an upper portion of the exposure lens. Alternatively, in order to inhibit damage to the surface of the exposure lens, the exposure lens may be implemented using a wear-resistant material having higher rigidity than other lenses in the lens unit.

Further, an outer diameter of each of the lenses **L1** and **L2** included in the first lens unit **110** may increase toward a lower portion (e.g., in a $-z$ -axis direction), but the embodiment is not limited thereto.

Light incident on the first lens unit **110** from the outside of the camera module **100** may pass through the liquid lens unit **140** and may be incident on the second lens unit **130**. The second lens unit **130** may be implemented as a single lens and may be implemented as two or more lenses that are aligned along the central axis to form an optical system. For example, as illustrated in FIG. 3A, the second lens unit **130** may include three lenses **L3**, **L4**, and **L5**, but this is exemplary, and two or less lenses or four or more lenses may be included in the second lens unit **130**.

Further, an outer diameter of each of the lenses **L3**, **L4**, and **L5** included in the second lens unit **130** may increase toward a lower portion thereof (e.g., in the $-z$ -axis direction), but the embodiment is not limited thereto.

In addition, unlike the liquid lens unit **140**, each of the first lens unit **110** and the second lens unit **130** may be a solid lens and may be formed of glass or plastic, but the embodiment is not limited to a specific material of each of the first lens unit **110** and the second lens unit **130**.

Further, the liquid lens unit **140** may include first to fifth regions **A1**, **A2**, **A3**, **A4**, and **A5**. Specifically, the fourth region **A4** and the fifth region **A5** are positioned on outermost sides of the camera module **100**, and the first region **A1**, the second region **A2**, and the third region **A3** may be disposed between the fourth region **A4** and the fifth region **A5**. In addition, the third region **A3** may be disposed between the first region **A1** and the second region **A2**. In addition, the first region **A1** may be disposed between the fourth region **A4** and the third region **A3**, and the second region **A2** may be disposed between the third region **A3** and the fifth region **A5**.

In other words, the first region **A1** and the second region **A2** may be regions disposed in openings of side surfaces of the holder **120**. The third region **A3** may be a region between the first region **A1** and the second region **A2**. In addition, the fourth region **A4** and the fifth region **A5** are regions that protrude from openings in the holder **120**, and are regions disposed outside the holder **120** at the openings.

Hereinafter, the first adhesive member **162** and the second adhesive member **164** will be described, and the liquid lens unit **140** will be described below.

The first adhesive member **162** may include first and second adhesive portions **162-1** and **162-2**, and the second adhesive member **164** may include third and fourth adhesive portions **164-1** and **164-2**. The adhesive portion may include an adhesive, an epoxy, or the like.

First, the first adhesive portion **162-1** couples the holder **120** and an upper surface **140TS** of the fourth region **A4** of the liquid lens unit **140**, and the third adhesive portion **164-1** couples the holder **120** and an upper surface of the fifth region **A5** of the liquid lens unit **140**. Here, the upper surface **140TS** of each of the fourth region **A4** and the fifth region **A5** of the liquid lens unit **140** is exemplified as being an upper surface of a first connection substrate **141**, but the embodiment is not limited thereto. For example, when the liquid lens unit **140** does not include connection substrates **141** and **144** or the spacer **143**, the upper surface of the liquid lens unit **140** may be an upper surface of a liquid lens **142**.

Further, the holder **120** may include a holder upper portion disposed above the liquid lens unit **140** (or the liquid lens **142**) and a holder lower portion disposed below the liquid lens unit **140** (or the liquid lens **142**). In addition, the holder **120** may also include a sidewall facing a side surface of the liquid lens **142** or the liquid lens unit **140**. Each of the first adhesive portion **162-1** and the third adhesive portion **164-1** may couple the holder upper region **120U** and the liquid lens unit **140**. In this way, each of the first adhesive portion **162-1** and the third adhesive portion **164-1** couples the holder **120** and the liquid lens unit **140** so that the liquid lens unit **140** may be fixed to the holder **120**.

Further, the second adhesive portion **162-2** may couple the holder **120** and a lower surface and a side surface of the fourth region **A4** of the liquid lens unit **140**. Here, the lower surface of the liquid lens unit **140** is exemplified as being a lower surface of a second connection substrate **144**, and the side surface of the liquid lens unit **140** is exemplified as being a side surface of the spacer **143**, but the embodiment is not limited thereto. For example, when the liquid lens unit **140** does not include the first and second connection substrates **141** and **144** or the spacer **143**, the lower surface and the side surface of the liquid lens unit **140** may be the lower surface and the side surface of the liquid lens **142**, respectively. Similarly, the fourth adhesive portion **164-2** may couple the holder **120** and a lower surface and a side surface of the fifth region **A5** of the liquid lens unit **140**. Here, the lower surface of the liquid lens unit **140** may be the lower surface of the second connection substrate **144**, and the side surface of the liquid lens unit **140** may be the side surface of the spacer **143**, but the embodiment is not limited thereto. For example, when the liquid lens unit **140** does not include the connection substrates **141** and **144** or the spacer **143**, the lower surface and the side surface of the liquid lens unit **140** may be the lower surface and the side surface of the liquid lens **142**, respectively.

In addition, each of the second adhesive portion **162-2** and the fourth adhesive portion **164-2** may couple the holder lower region **120D** and the liquid lens unit **140**. In this way, each of the second adhesive portion **162-2** and the fourth adhesive portion **164-2** may couple the holder **120** and the liquid lens unit **140** so that the opening of the holder **120** may be sealed. For example, the first adhesive portion **162-1** and the second adhesive portion **162-2** may be connected to each other, and the third adhesive portion **164-1** and the fourth adhesive portion **164-2** may be connected to each other, thereby performing the above-described sealing.

Although not shown in the drawings, the third adhesive member **166** may be disposed to fill a separation space (or gap) between the upper surface of the holder **120** and the first cover **170**. In addition, the third adhesive member **166** may be omitted in some cases. In this case, the separation space between the upper surface of the holder **120** and the first cover **170** may be empty.

The embodiment is not limited to the shapes of the first to third adhesive members **162**, **164**, and **166** described above. That is, the first to third adhesive members **162**, **164**, and **166** may have various shapes as long as the adhesive members seal the inside of the holder **120** in order to inhibit an inflow of external foreign substances into the holder **120** through the opening of the holder **120**. For example, when the third adhesive member **166** is disposed to fill the separation space between the upper surface of the holder **120** and the first cover **170** in a state in which the first adhesive member **162** and the second adhesive member **164** are disposed to seal the openings of the holder **120**, the third region A3 of the liquid lens unit **140** disposed in an inner space of the holder **120** may be sealed from the outside. Accordingly, in the camera module **100** according to the embodiment, reliability against foreign substances can be improved, optical performance degradation can be inhibited, and a defect rate can be reduced.

In addition, each of the first to fourth adhesive portions **162-1**, **164-1**, **162-2**, and **164-2** may have various shapes. That is, the first adhesive portion **162-1**, the second adhesive portion **162-2**, the third adhesive portion **164-1**, and the fourth adhesive portion **164-2** may have shapes corresponding to the shapes of the openings of the holder **120**.

In addition, an adhesive may be disposed in the first and second regions A1 and A2 as well as the fourth region A4 and the fifth region A5, but the present invention is not limited thereto.

Further, the above-described first cover **170** may be disposed to surround the holder **120**, the liquid lens unit **140**, the middle base **172**, and a sensor base **178**. Accordingly, the first cover **170** may be in contact with a shoulder-side upper surface **120S** of the holder **120**. In this case, when an upper surface **162S**, **164S** of each of the first adhesive portion **162-1** and the third adhesive portion **164-1** is positioned higher than the shoulder-side upper surface **120S** of the holder **120**, the first cover **170** may be in contact with the upper surface **162S**, **164S** of each of the first adhesive portion **162-1** and the third adhesive portion **164-1**, instead of the shoulder-side upper surface **120S** of the holder **120**. Accordingly, the first cover **170** may be unstably fixed to the holder **120**. In order to inhibit this, the upper surface **120S** of the holder **120** may be disposed higher than the upper surface **162S** of the first adhesive portion **162-1** by a predetermined height. Similarly, the upper surface **120S** of the holder **120** may be positioned higher than the upper surface **164S** of the third adhesive portion **164-1** by a predetermined height.

Further, the height of the upper surface **120S** of the holder **120** may be equal to the height of the upper surface **162S** of the first adhesive portion **162-1**, and the height of the upper surface **120S** of the holder **120** may be equal to the height of the upper surface **164S** of the third adhesive portion **164-1**.

In addition, in the above description, the liquid lens unit **140**, which is given when describing the features of the first adhesive member **162** and the second adhesive member **164**, includes the first and second connection substrates **141** and **144**. However, the liquid lens unit **140**, which is given when describing the features of the first adhesive member **162** and

the second adhesive member **164** described above, may not include the first and second connection substrates **141** and **144**.

Further, the first cover **170** may be disposed to surround the holder **120**, the liquid lens unit **140**, the third adhesive member **166**, and the middle base **172** and may protect these (the holder **120**, the liquid lens unit **140**, the third adhesive member **166**, and the middle base **172**) from an external impact. In particular, as the first cover **170** is disposed, the plurality of lenses, which form an optical system, may be protected from an external impact.

Further, the first cover **170** may include an upper opening **170H** formed in an upper surface of the first cover **170**. Accordingly, the first lens unit **110** disposed in the holder **120** may be exposed to external light.

In addition, the middle base **172** may be disposed to surround a hole of the holder **120**. Accordingly, the middle base **172** may include an accommodation hole **172H** for accommodating the hole of the holder **120**. An inner diameter of the middle base **172** (i.e., a diameter of the accommodation hole **172H**) may be greater than or equal to an outer diameter of the hole of the holder **120**. In addition, each of the accommodation hole **172H** of the middle base **172** and the hole of the holder **120** is illustrated as having a circular shape, but the embodiment is not limited thereto, and these holes may be changed to various shapes. In addition, the middle base **172** may be mounted on the main substrate **150** to be spaced apart from a circuit element **151** on the main substrate **150**.

Like the upper opening **170H** of the first cover **170**, the accommodation hole **172H** may be formed at a position near a center of the middle base **172** at a position corresponding to the position of the image sensor **30** disposed on the camera module **100**.

Further, the camera module **100** may further include the sensor base **178** and a filter **176** and may further include a circuit cover **154**.

The filter **176** may filter light passing through the first lens unit **110**, the liquid lens unit **140**, and the second lens unit **130** in a specific wavelength range. The filter **176** may be an infrared (IR) light-blocking filter or an ultraviolet (UV) light-blocking filter, but the embodiment is not limited thereto.

Further, the filter **176** may be disposed above the image sensor **30**. The filter **176** may be disposed inside the sensor base **178**. For example, the filter **176** may be disposed or mounted in an inner recess or on a stepped portion of the sensor base **178**.

The sensor base **178** may be disposed below the middle base **172** and attached to the main substrate **150**. The sensor base **178** may surround the image sensor **30** and protect the image sensor **30** from external foreign substances or impacts.

Next, the main substrate **150** may be disposed below the middle base **172** and may include a recess in which the image sensor **30** may be mounted, seated, contacted, fixed, provisionally fixed, supported, coupled, or accommodated, the circuit element **151**, a connection part (or a flexible printed circuit board (FPCB)) **152**, and a connector **153**.

Specifically, the main substrate **150** may include a holder region in which the holder **120** is disposed, and an element region in which a plurality of circuit elements **151** are disposed.

The main substrate **150** may be implemented as a rigid-flexible printed circuit board (RFPCB) including the FPCB **152**. The FPCB **152** may be bent based on the requirement of a space in which the camera module **100** is mounted.

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In addition, the circuit element **151** of the main substrate **150** may configure a control module configured to control the liquid lens unit **140** and the image sensor **30**. Here, the control module will be described below.

In addition, the circuit element **151** may include at least one of a passive element and an active element and may have various widths and heights. The circuit element **151** may be provided as a plurality of circuit elements **151** and may protrude outward with a height greater than a height of the main substrate **150**. The plurality of circuit elements **151** may be disposed not to overlap the holder **120** in a direction parallel to the optical axis LX. For example, the plurality of circuit elements **151** may include a power inductor, a gyro sensor, and the like, but the embodiment is not limited to a specific type of the circuit element **151**.

The circuit cover **154** may be disposed to cover the circuit element **151**. Accordingly, the circuit cover **154** may protect the circuit element **151** disposed on an upper portion of the main substrate **150** from an external impact. Further, to this end, the circuit cover **154** may include an accommodation space for accommodating and covering the circuit element **151** in consideration of the shape and position of the circuit element **151** disposed on the main substrate **150**. In addition, the circuit cover **154** may have an electromagnetic shielding function.

The image sensor **30** may perform a function of converting light that has passed through the first lens unit **110**, the liquid lens unit **140**, and the second lens unit **130** of the lens assembly (**110**, **120**, **130**, **140**, **162**, and **164**) into image data. More specifically, the image sensor **30** may generate image data by converting light into an analog signal through a pixel array including a plurality of pixels and synthesizing a digital signal corresponding to the analog signal.

Referring to FIGS. **3** and **4**, the liquid lens unit **140** may include the first connection substrate (or individual electrode connection substrate) **141**, the liquid lens (or liquid lens body) **142**, the spacer **143**, and the second connection substrate (or common electrode connection substrate) **144**.

The first connection substrate **141** may electrically connect a plurality of first electrodes (not shown) included in the liquid lens **142** to the main substrate **150** and may be disposed above the liquid lens **142**. The first connection substrate **141** may be implemented as an FPCB.

Further, the first connection substrate **141** may be electrically connected to an electrode pad **150-1**, which is formed on the main substrate **150**, through a connection pad **141-1** electrically connected to each of the plurality of first electrodes. To this end, after the liquid lens unit **140** is inserted into the inner space of the holder **120**, the first connection substrate **141** may be bent in the $-z$ -axis direction toward the main substrate **150**, and then, the connection pad **141-1** and the electrode pad **150-1** may be electrically connected to each other through a conductive epoxy. In another embodiment, the first connection substrate **141** may be connected to a first holder surface electrode, which is conductive and is disposed, formed, or coated on a surface of the holder **120**, to be electrically connected to the main substrate **150** through the first holder surface electrode, which is conductive and is disposed on the surface of the holder **120**, but the embodiment is not limited thereto.

The second connection substrate **144** may electrically connect a second electrode (not shown) included in the liquid lens **142** to the main substrate **150** and may be disposed below the liquid lens **142**. The second connection substrate **144** may be implemented as an FPCB or a single metal substrate (conductive metal plate).

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Further, the second connection substrate **144** may be electrically connected to an electrode pad, which is formed on the main substrate **150**, through a connection pad electrically connected to the second electrode. To this end, after the liquid lens unit **140** is inserted into the inner space of the holder **120**, the second connection substrate **144** may be bent in the $-z$ -axis direction toward the main substrate **150**. In another embodiment, the second connection substrate **144** may be connected to a surface electrode, which is conductive and is disposed, formed, or coated on the surface of the holder **120**, to be electrically connected to the main substrate **150** through the surface electrode, which is conductive and is disposed on the surface of the holder **120**, but the embodiment is not limited thereto.

The liquid lens **142** may include a cavity CA. In addition, an opening area in a direction in which light enters the cavity CA may be smaller than an opening area in an opposite direction. Alternatively, the liquid lens **142** may be disposed such that a direction of inclination of the cavity CA is opposite to that illustrated in the drawing. In addition, the opening area in the direction in which light enters the cavity CA may be greater than the opening area in the opposite direction. In addition, when the liquid lens **142** is disposed such that the direction of inclination of the cavity CA is reversed, according to the direction of inclination of the liquid lens **142**, a portion or the entirety of the arrangement of components included in the liquid lens **142** may be changed, or only the direction of inclination of the cavity CA may be changed and the remainder of the arrangement of the components may not be changed.

The spacer **143** may be disposed to surround the liquid lens **142** and may protect the liquid lens **142** from an external impact. To this end, the spacer **143** may have a shape in which the liquid lens **142** may be mounted in, seated in, in contact with, fixed to, provisionally fixed to, supported by, coupled to, or disposed in the spacer **143**.

For example, the spacer **143** may include a hollow space **143H** in which the liquid lens **142** is accommodated, and a frame configured to surround the hollow space **143H** formed in a center thereof. In this way, the spacer **143** may have a centrally-hollowed quadrangular planar shape (hereinafter, referred to as a “ z ”-shaped form), but the embodiment is not limited thereto.

Further, the spacer **143** may be disposed between the first connection substrate **141** and the second connection substrate **144** and may be disposed to protrude from the opening of the holder **120**.

Further, the liquid lens **142** may include a plurality of different types of liquids LQ1 and LQ2, first to third plates **147**, **145**, and **146**, first and second electrodes E1 and E2, and an insulating layer **148**. The liquid lens **142** may further include the optical layer **190**.

In addition, the plurality of liquids LQ1 and LQ2 may be accommodated in the cavity CA, and may include a first liquid LQ1 having conductivity and a second liquid (or an insulating liquid) LQ2 having non-conductivity. Further, the first liquid LQ1 and the second liquid LQ2 may not be mixed with each other, and an interface BO may be formed at a contact portion between the first and second liquids LQ1 and LQ2. In addition, the second liquid LQ2 may be disposed on the first liquid LQ1, but the embodiment is not limited thereto.

Further, in a cross-sectional shape of the liquid lens **142**, an edge of each of the first and second liquids LQ1 and LQ2 may be thinner than a central portion thereof. However, the present invention is not limited thereto.

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An inner side surface of the first plate 147 may form a sidewall i of the cavity CA. The first plate 147 may include upper and lower openings having a predetermined inclined surface. That is, the cavity CA may be a region surrounded by the inclined surface of the first plate 147, the second plate 145, and the third plate 146.

Further, according to the embodiment, a size (or an area or a width) O2 of the lower opening may be larger than a size (or an area or a width) O1 of the upper opening. Here, the size of each of the upper and lower openings may be a sectional area in a horizontal direction (e.g., an x-axis direction and a y-axis direction). For example, the size of the opening may be a radius when the opening has a circular cross section and may be a diagonal length when the opening has a square cross section. In addition, a diameter of the opening may be changed depending on a field of view (FOV) required for the liquid lens 142 or the role of the liquid lens 142 to be performed in the camera module 100.

In addition, each opening may have the form of a hole having a circular cross section, and an inclined surface thereof may have an inclination. The interface BO may move along the inclined surface of the cavity CA by a driving voltage.

In addition, as described above, the first liquid LQ1 and the second liquid LQ2 may be filled, accommodated, or disposed in the cavity CA. In addition, the cavity CA is a portion through which the light that has passed through the first lens unit 110 is transmitted. Further, since the first plate 147 is positioned outside the cavity CA, the first plate 147 may be made of a transparent material. In addition, the first plate 147 may include impurities so that light is not easily transmitted therethrough.

In addition, the electrode may be disposed on each of one surface and the other surface of the first plate 147. A plurality of first electrodes E1 may be disposed to be spaced apart from the second electrode E2 and disposed on one surface (e.g., an upper surface, a side surface, or a lower surface) of the first plate 147. The second electrode E2 may be disposed on at least a partial region of the other surface (e.g., a lower surface) of the first plate 147 and may be in direct contact with the first liquid LQ1.

Further, the first electrodes E1 may include n electrodes (hereinafter, referred to as "individual electrodes"), and the second electrode E2 may include a single electrode (hereinafter, referred to as a "common electrode"). Here, n may be an integer greater than or equal to 2. Here, a case of four first electrodes E1 and one second electrode E2 will be described below. That is, both ends electrically connected to the liquid lens 142 may be any one of the plurality of first electrodes E1 and the second electrode E2.

Further, a portion of the second electrode E2 (i.e., an electrode sector of the second electrode E2) disposed on the other surface of the first plate 147 may be exposed to the first liquid LQ1 having conductivity.

Each of the first and second electrodes E1 and E2 may be made of a conductive material. For example, the first electrode E1 and the second electrode E2 may be made of a metal.

Further, the second plate 145 may be disposed on one surface of the first electrode E1. That is, the second plate 145 may be disposed above the first plate 147. Specifically, the second plate 145 may be disposed above an upper surface of the first electrode E1 and the cavity CA.

The third plate 146 may be disposed on one surface of the second electrode E2. That is, the third plate 146 may be disposed below the first plate 147. Specifically, the third

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plate 146 may be disposed below a lower surface of the second electrode E2 and the cavity CA.

The second plate 145 and the third plate 146 may be disposed opposite to each other with the first plate 147 interposed therebetween. In addition, at least one of the second plate 145 and the third plate 146 may be omitted.

At least one of the second and third plates 145 and 146 may have a quadrangular planar shape. The third plate 146 may be brought into contact with and bonded to the first plate 147 in a bonding region thereof around the edge.

The second plate 145 and the third plate 146 are regions through which light passes, and may be made of a transparent material. For example, each of the second and third plates 145 and 146 may be made of glass, and may be formed of the same material for convenience of the process. In addition, the edge of each of the second and third plates 145 and 146 may have a quadrangular shape, but the present invention is not necessarily limited thereto.

In addition, the second plate 145 may allow light incident from the first lens unit 110 to enter the cavity CA of the first plate 147.

Further, the third plate 146 may allow the light that has passed through the cavity CA of the first plate 147 to travel to the second lens unit 130. The third plate 146 may be in direct contact with the first liquid LQ1.

According to the embodiment, the third plate 146 may have a diameter larger than a diameter of a wider opening among the upper and lower openings of the first plate 147. In addition, the third plate 146 may include a peripheral region spaced apart from the first plate 147.

For example, an actual effective lens region of the liquid lens 142 may be narrower than the diameter (e.g., O2) of the wider opening among the upper and lower openings of the first plate 147. That is, when a region in a small radius about a central portion of the liquid lens 142 is used as an actual path for transmitting light, a diameter O3 of a central region of the third plate 146 may be smaller than a diameter (e.g., O2) of a wider opening among third and fourth openings of the first plate 147.

The insulating layer 148 may be disposed to cover a portion of a lower surface of the second plate 145 in an upper region of the cavity CA. That is, the insulating layer 148 may be disposed between the second liquid LQ2 and the second plate 145.

Further, the insulating layer 148 may be disposed to cover a portion of the first electrodes E1 that forms a sidewall of the cavity CA. In addition, the insulating layer 148 may be disposed on a lower surface of the first plate 147 to cover a portion of the first electrode E1, the first plate 147, and the second electrode E2. Accordingly, the insulating layer 148 may block the first electrode E1 from coming into contact with the first liquid LQ1 and block the first electrode E1 from coming into contact with the second liquid LQ2.

The insulating layer 148 may cover one electrode (e.g., the first electrode E1) of the first and second electrodes E1 and E2 and may expose a portion of another electrode (e.g., the second electrode E2) so that electric energy is applied to the first liquid LQ1 having conductivity.

Meanwhile, the optical layer 190 may be disposed on one surface of at least one of the second plate 145 and the third plate 146. For example, the optical layer 190 may be disposed on at least one of upper and lower portions of the second plate 145, may be disposed on at least one of upper and lower portions of the third plate 146, or may be disposed on at least one of upper and lower portions of each of the second and third plates 145 and 146.

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In addition, the liquid lens **142** and the optical layer **190** are illustrated separately from each other, but the optical layer **190** may be a component of the liquid lens **142**. In addition, the optical layer **190** is illustrated as being a single layer, but this is only to indicate the presence of the optical layer **190**. That is, the optical layer **190** may be a single layer or a multilayer.

In addition, the optical layer **190** may include at least one of an ultraviolet light-blocking layer, an anti-reflection layer, or an infrared light-blocking layer. The optical layer **190** may be disposed to overlap the image sensor **30** in a direction of the optical axis LX (i.e., a z-axis) or in a direction parallel to the optical-axis direction. In addition, the optical layer **190** may be disposed in the third region A3 of the liquid lens unit **140**. In addition, the anti-reflection layer may serve to inhibit light from being reflected from the second plate **145** or the third plate **146**, may reduce a decrease in light transmittance due to Fresnel loss in the liquid lens **142**, and may inhibit a decrease in visibility of the liquid lens **142** at night. In particular, although not shown in the drawings, the anti-reflection layer may be disposed on an inclined surface and a lower surface of the insulating layer **148**, and may inhibit a decrease in the quality of light transmitted to the image sensor **30** due to the reflection of light.

In addition, the infrared light-blocking (IR cut-off) layer may block light in an infrared region. The infrared light-blocking layer may remove hot spots from an image by inhibiting external infrared light from being incident on the liquid lens **142**, and may inhibit a decrease in visibility at night by reducing the reflection of light from a surface of the liquid lens **142**.

In addition, at least one of the ultraviolet light-blocking layer, the anti-reflection layer, and the infrared light-blocking layer may be disposed on at least one of a light-receiving portion and a light-emitting portion of the camera module **100** according to the embodiment.

In addition, according to the embodiment, the optical layer **190** may have a coated form or a film form. For example, the anti-reflection layer of the optical layer **190** may be formed by coating at a low temperature by a spray method or the like.

FIGS. 5A to 5E are views for describing a method of driving the liquid lens unit. In the following description, a voltage will be described as being applied between the first electrode and the second electrode.

First, in FIG. 5A, a first voltage V1 is applied to the liquid lens unit **140** so that an interface BO1 of the liquid lens may be convex in a light traveling direction. Thus, it is illustrated that the liquid lens unit operates as a concave lens. In addition, in FIG. 5B, a second voltage V2 greater than the first voltage V1 is applied to the liquid lens unit **140** so that an interface BO2 of the liquid lens may be perpendicular to the light traveling direction. Thus, it is illustrated that the liquid lens unit does not change the light traveling direction. In addition, in FIG. 5C, a third voltage V3 greater than the second voltage V2 is applied to the liquid lens unit **140** so that an interface BO3 of the liquid lens may be convex in a direction opposite to the light traveling direction. Thus, it is illustrated that the liquid lens unit operates as a convex lens. In this case, when the first to third voltages V1 to V3 are applied, the interfaces BO1, BO2, and BO3 of the liquid lens in the liquid lens unit **140** may have different curvatures.

That is, in the liquid lens unit **140** according to the embodiment, it is illustrated that the curvature or diopter of the liquid lens of the liquid lens unit is changed depending on the level of the applied voltage, but the present invention

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is not limited thereto, and the curvature or diopter of the liquid lens may also be changed depending on a pulse width of an applied pulse.

In addition, FIG. 5D illustrates that the liquid lens in the liquid lens unit **140** operates as a convex lens as the liquid lens in the liquid lens unit **140** has an interface the same as the interface BO3 as in FIG. 5C. Thus, according to FIG. 5D, incident light LPa is converged, and corresponding output light LPb is output.

In addition, FIG. 5E illustrates that the liquid lens in the liquid lens unit **140** has an asymmetric curved surface (e.g., an upper portion of the interface is convex in a direction opposite to the light traveling direction), so that the light traveling direction is changed to one side (e.g., an upper side). That is, according to FIG. 5D, the incident light LPa is converged to the upper side, and corresponding output light LPc is output.

FIG. 6 is a view for describing the liquid lens unit in which the interface is adjusted according to a driving voltage.

Referring to FIG. 6, FIG. 6A illustrates the liquid lens unit, and FIG. 6B illustrates an equivalent circuit of the liquid lens unit.

Referring to FIG. 6A, a lens **142** whose focal length is adjusted according to a driving voltage may receive the voltage through individual terminals L1, L2, L3, and L4 that are disposed in four different directions to have the same angular distance therebetween. The individual terminals may be disposed with the same angular distance therebetween with respect to a central axis of the liquid lens and may include four individual terminals. Each of the four individual terminals may be disposed at one of four corners of the liquid lens. When a voltage is applied through the individual terminals L1, L2, L3, and L4, an interface of the liquid lens may be deformed by a driving voltage, which is formed due to interaction between the applied voltage and a voltage applied to a common terminal C0, which will be described below.

Further, referring to FIG. 6B, the liquid lens **142** may have one side to which an operating voltage is applied from the different individual terminals L1, L2, L3, and L4 and another side electrically connected to the common terminal C0. In addition, the common terminal C0 may be connected to a plurality of capacitors **149**. In addition, the plurality of capacitors **149** included in the equivalent circuit may have a small capacitance of about tens to about 200 pico-farads (pF) or less. The terminals of the liquid lens may be referred to as electrode sectors or sub-electrodes.

FIGS. 7A to 7C are views illustrating a structure of the liquid lens unit,

Referring to FIG. 7A, the liquid lens unit **140** may include a common electrode E2 (corresponds to the second electrode, and hereinafter, the second electrode and the common terminal are used interchangeably) at one side thereof. In this case, the common electrode E2 may be disposed in the form of a tube, and a liquid LQ may be disposed in a lower region of the common electrode E2, in particular, in a region corresponding to a hollow.

Meanwhile, although not shown in the drawing, in order to insulate the common electrode E2, an insulating layer may be disposed between the common electrode E2 and the liquid.

In addition, as shown in FIG. 7B, a plurality of first electrodes E11 to E14 (the first electrodes correspond to the individual electrodes, and hereinafter, the first electrodes and the individual terminals are used interchangeably) may be disposed below the common electrode, in particular,

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below the liquid LQ. The plurality of first electrodes E11 to E14, in particular, may be disposed in a shape surrounding the liquid LQ.

In addition, a plurality of insulating layers 148a to 148d for insulation may be respectively disposed between the plurality of first electrodes E11 to E14 and the liquid LQ.

Referring to FIG. 7C, the liquid lens unit 140 may include the plurality of first electrodes E11, E12, E13, and E14 disposed on the first connection substrate 141, the plurality of insulating layers 148a to 148d for insulating the first electrodes E11, E12, E13, and E14, the second liquid LQ2 disposed on the first electrodes E11, E12, E13, and E14, the first liquid LQ1 disposed on the second liquid LQ2, the second electrode E2 disposed apart from the second liquid LQ2 and the first electrodes E11, E12, E13, and E14, and the second connection substrate 144 disposed on the second electrode E2. In other configurations, the above-described contents may be applied.

Further, the second electrode E2 may be formed in the form of a tube with a hollow. In addition, the second liquid LQ2 and the first liquid LQ1 may be disposed in a hollow region. The second liquid LQ2 may be disposed in a circular shape as shown in FIGS. 7A and 7B.

Meanwhile, the hollow region may have a size that increases in a direction from bottom to top, and accordingly, each of the first electrodes E11, E12, E13, and E14 may have a size that decreases in a direction from bottom to top.

In FIG. 7C, a first-first electrode E11 and a first-second electrode E12 among the first electrodes E11, E12, E13, and E14 are each illustrated as being formed to be inclined and having a size that decreases in a direction from bottom to top. Meanwhile, unlike FIGS. 7A to 7C, the first electrodes E11, E12, E13, and E14 may be positioned above the second electrode E2. In other words, the first electrodes E11, E12, E13, and E14 may be positioned below the second electrode E2, but the present invention is not limited thereto.

Further, the first electrodes E11, E12, E13, and E14 are not limited to the above-described number and may be formed in various numbers.

Further, when a pulse-type electrical signal is applied to the first-first electrode E11 and the first-second electrode E12 after the pulse-type electrical signal is applied to the second electrode E2 and a predetermined period of time elapses, a potential difference is generated between the second electrode E2, the first-first electrode E11, and the first-second electrode E12, and accordingly, a shape of the first liquid LQ1 having electrical conductivity may be changed, and a shape of the second liquid LQ2 inside the second liquid LQ2 may be changed according to the change in the shape of the first liquid LQ1.

Meanwhile, in the present invention, proposed is a method of simply and quickly detecting the curvature of the second liquid LQ2 that changes according to an electrical signal applied to each of the first electrodes E11, E12, E13, and E14 and the second electrode E2. To this end, a sensor unit of the present invention may detect a size of an area or a change in the area of a boundary region Ac0 between the first liquid LQ1 and the first insulator 148a on the first electrode E11 in the liquid lens unit 140.

In FIG. 7C, AM0 is exemplarily given as an area of the boundary region Ac0. In particular, it is illustrated that the area of the boundary region Ac0, which is in contact with the first liquid LQ1, of an inclined portion of a first insulating layer 148a on the first-first electrode E11 is AM0. In addition, it is illustrated that the second liquid LQ2 is not

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concave or convex and is parallel to the first connection substrate 141 and the like. The curvature at this point may be defined as 0, for example.

For the boundary region Ac0, which is in contact with the first liquid LQ1, of the inclined portion of the first insulating layer 148a on the first-first electrode E11, a capacitance C may be formed according to Equation 1.

$$C = \epsilon \frac{A}{d} \quad \text{Equation 1}$$

Here, ϵ denotes a dielectric constant of a dielectric 148a, A denotes the area of the boundary region Ac0, and d denotes a thickness of the first dielectric 148a. The capacitance C may be a capacitance of the capacitor 149 illustrated in FIG. 6. Accordingly, since the above-described sensor unit detects the capacitance of the capacitor 149, the sensor unit may include the capacitor 149.

In addition, when it is assumed here that F and d are fixed values, the area of the boundary region Ac0 may greatly affect the capacitance C. That is, as the area of the boundary region Ac0 increases, the capacitance C formed in the boundary region Ac0 may increase. Meanwhile, since the area of the boundary region Ac0 is varied as the curvature of the second liquid LQ2 is varied, the area of the boundary region Ac0 or the capacitance C formed in the boundary region Ac0 may be detected using the sensor unit.

Meanwhile, in the present specification, the capacitance in FIG. 7C may be defined as CAc0.

FIGS. 8A to 8E are views for describing a variation in the lens curvature of the liquid lens unit.

FIGS. 8A to 8E illustrate that a first curvature BOa is formed in the second liquid LQ2 according to the application of an electrical signal to each of the first electrodes E11, E12, E13, and E14 and the second electrode E2.

In FIG. 8A, it is illustrated that an area of a boundary region Aaa is AMa (>AM0) when the first curvature BOa is formed in the second liquid LQ2. In particular, it is illustrated that the area of the boundary region Aaa, which is in contact with an electroconductive aqueous solution LQ1, of the inclined portion of the first insulating layer 148a on the first-first electrode E11 is AMa.

According to Equation 1, the area of the boundary region Aaa in FIG. 8A is larger than that in FIG. 7C, and thus the capacitance of the boundary region Aaa becomes larger. Meanwhile, the capacitance in FIG. 8A may be defined as CAaa, which is greater than CAc0 that is the capacitance in FIG. 7C. In addition, the first curvature BOa at this point may be defined as having a value of positive polarity. For example, the first curvature BOa may be defined as having a level of +2.

FIG. 8B illustrates that a second curvature Bob is formed in the second liquid LQ2 according to application of an electrical signal to each of the first electrodes E11, E12, E13, and E14 and the second electrode E2.

Referring to FIG. 8B, it is illustrated that an area of a boundary region Aba is AMb (>AMa) when the second curvature Bob is formed in the second liquid LQ2. In particular, it is illustrated that the area of the boundary region Aba, which is in contact with the first liquid LQ1, of the inclined portion of the first insulating layer 148a on the first-first electrode E11 is AMb.

According to Equation 1, the area of the boundary region Aba in FIG. 8B is larger than that in FIG. 8A, and thus the capacitance of the boundary region Aba becomes larger.

Meanwhile, the capacitance in FIG. 8B may be defined as CAba, which is greater than CAaa that is the capacitance in FIG. 8A.

The second curvature BOb at this point may be defined as having a value of positive polarity greater than the first curvature BOa. For example, the second curvature BOb may be defined as having a level of +4.

Meanwhile, according to FIGS. 8A and 8B, the liquid lens unit 140 operates as a convex lens, and accordingly, output light LP1a formed by converging incident light LP1 is output.

Next, FIG. 8C illustrates that a third curvature BOc is formed in the second liquid LQ2 according to application of an electrical signal to each of the first electrodes E11, E12, E13, and E14 and the second electrode E2.

In particular, in FIG. 8C, it is illustrated that an area of a left boundary region Aca is AMa and an area of a right boundary region Acb is AMa (>AMa).

In particular, it is illustrated that the area of the boundary region Aca, which is in contact with the first liquid LQ1, of the inclined portion of the first insulating layer 148a on the first-first electrode E11 is AMa, and the area of the boundary region Acb, which is in contact with the first liquid LQ1, of the inclined portion of the second insulator 148b on the second electrode E12 is AMb.

Accordingly, the capacitance of the left boundary region Aca may be CAaa, and the capacitance of the right boundary region Acb may be CAba. The third curvature BOc at this point may be defined as having a value of positive polarity. For example, the third curvature BOc may be defined as having a level of +3. Meanwhile, according to FIG. 8C, the liquid lens unit 140 operates as a convex lens, and accordingly, output light LP1b formed by converging incident light LP1 further to one side may be output.

Next, FIG. 8D illustrates that a fourth curvature BOd is formed in the second liquid LQ2 according to application of an electrical signal to each of the first electrodes E11, E12, E13, and E14 and the second electrode E2.

In FIG. 8D, it is illustrated that an area of a boundary region Ada is AMd (<AM0) when the fourth curvature BOd is formed in the second liquid LQ2. In particular, it is illustrated that the area of the boundary region Ada, which is in contact with the first liquid LQ1, of the inclined portion of the first insulating layer 148a on the first-first electrode E11 is AMd.

According to Equation 1, the area of the boundary region Ada in FIG. 8D is smaller than that in FIG. 8C, and thus the capacitance of the boundary region Ada becomes smaller. Meanwhile, the capacitance in FIG. 8D may be defined as CAda, which is smaller than CAC0 that is the capacitance in FIG. 7C.

The fourth curvature BOd at this point may be defined as having a value of negative polarity. For example, the fourth curvature BOd may be defined as having a level of -2.

Next, FIG. 8E illustrates that a fifth curvature BOe is formed in the second liquid LQ2 according to application of an electrical signal to each of the first electrodes E11, E12, E13, and E14 and the second electrode E2.

In FIG. 8E, it is illustrated that an area of a boundary region Aea is AMe (<AMd) when the fifth curvature BOe is formed in the second liquid LQ2. In particular, it is illustrated that the area of the boundary region Aea, which is in contact with the first liquid LQ1, of the inclined portion of the first insulating layer 148a on the first-first electrode E11 is AMe.

According to Equation 1, the area of the boundary region Aea in FIG. 8E is smaller than that in FIG. 8D, and thus the

capacitance of the boundary region Aea becomes smaller. Meanwhile, the capacitance in FIG. 8E may be defined as CAea, which is smaller than CAda that is the capacitance in FIG. 8D. The fifth curvature BOe at this point may be defined as having a value of negative polarity. For example, the fifth curvature BOe may be defined as having a level of -4.

Meanwhile, according to FIGS. 8D and 8E, the liquid lens unit 140 operates as a concave lens, and accordingly, output light LP1c formed by diverging incident light LP1 is output.

FIG. 9 is a block diagram of a camera module according to an embodiment, FIG. 10 illustrates a modified example of FIG. 9, and FIGS. 11 to 13 are views referenced in the description of FIGS. 9 and 10.

First, a term "unit" used herein includes a software, a field-programmable gate array (FPGA), or a hardware component such as an application-specific integrated circuit (ASIC), and the "unit" performs certain functions. However, the "unit" is not limited to the software or hardware. The "unit" may be configured to reside on an addressable storage medium or may be configured to play back one or more processors. Therefore, the "unit" may include, for example, components (such as software components, object-oriented software components, class components, and task components), processes, functions, attributes, procedures, subroutines, segments of program code, drivers, firmware, microcodes, circuitry, data, databases, data structures, tables, arrays, and variables. Components and functions provided in "units" may be combined to be a smaller number of components and "units," or may be further divided into additional components and "units." Furthermore, the components and "units" may be implemented to play back one or more central processing units (CPUs) in a device or a secure multimedia card.

Referring to FIG. 9, a camera module 200 may include a control circuit 210, a lens assembly 220, and a gyro sensor 230. The control circuit 210 may correspond to the aforementioned control circuit 20, and the lens assembly 220 may correspond to the aforementioned lens assembly 10, i.e., the lens assembly (110, 120, 130, 140, 162, and 164). In addition, the camera module 200 may further include an image sensor 30 and an image processor 31.

First, the gyro sensor 230 may detect an angular velocity of a liquid lens 280 or the camera module, and may also be positioned in a controller 240. That is, the position of the gyro sensor is not limited to such a position. In addition, the gyro sensor 230 may detect an angular velocity (or an angle) of movement in two directions, for example, a yaw-axis direction and a pitch-axis direction, in order to compensate for hand tremor of the optical device in an up-down direction and a left-right direction. The gyro sensor 230 may generate a motion signal having information on the detected angular velocity and provide the motion signal to the voltage controller 240. Hereinafter, descriptions will be made using X⁺, X⁻, Y⁺, and Y⁻.

In addition, the image sensor 30 may convert light passing through the liquid lens 280 into an electrical signal. Further, the image processor 31 may perform image processing on the basis of the electrical signal from the image sensor 30. It should be understood that the image processor 31 may be positioned in the control circuit 210 or the camera module according to the embodiment or an external system (e.g., a terminal).

An external system 40 may be positioned outside the camera module. For example, the external system 40 may be a terminal and may provide the electrical signal to the camera module so that an interface of the liquid lens 280 in

the camera module has a desired diopter. In an embodiment, the external system **40** may provide an electrical signal for setting a focus through the image processor **31** to the camera module (e.g., the control circuit **210**). The external system **40** may transmit a signal, which is a digital code, to the controller **240**. As described above, the external system **40** may be positioned outside the camera module. For example, the external system **40** may be positioned on the terminal and may further include a control unit, a controller, and the like. In addition, the external system **40** may be connected to the camera module by an inter-integrated circuit (I2C) communication method. However, the present invention is not limited thereto, and the connection may be made by other communication methods other than the I2C communication method.

The control circuit **210** may include the controller **240**, a lens driving unit **250**, and a power supply unit **260** and may control the operation of the lens assembly **220** including the liquid lens **280**.

The controller **240** may have a configuration for performing an auto focus (AF) function and an optical image stabilization (OIS) function. The controller **240** may control the liquid lens **280** included in the lens assembly **220** using a user request or a detected result (e.g., a detection signal of the gyro sensor **230**). Here, the liquid lens **280** may correspond to the liquid lens unit described above.

The controller **240** may calculate a driving voltage corresponding to a desired shape of the liquid lens **280**. Specifically, the controller **240** may receive information (i.e., information on a distance to an object) for the AF function from an optical device, the internal component of the camera module **200** (e.g., the image sensor **30**), or the external component (e.g., including a distance sensor or an application processor), and may calculate the driving voltage corresponding to the desired shape of the liquid lens **280** on the basis of a focal length, which is for focusing on the object, through the distance information.

In addition, the controller **240** may obtain a driving voltage code corresponding to the calculated driving voltage by referring to a driving voltage table and output the obtained driving voltage code to the lens assembly **220** (e.g., a driving voltage providing unit (not shown)). In this case, the controller **240** may have a driving voltage table in which the driving voltage code is mapped to generate the driving voltage. Alternatively, the controller **240** may further include a driving voltage unit (not shown) configured to generate the driving voltage, and the driving voltage unit may be positioned in a liquid lens module or in the camera module.

Alternatively, the controller **240** may generate an analog-type driving voltage corresponding to a provided digital-type driving voltage code on the basis of the digital-type driving voltage code and may provide the analog-type driving voltage to the lens assembly **220**. Thus, the position of the driving voltage providing unit may be changed.

In an embodiment, the controller **240** may output a driving signal, which is a voltage code in a data form (e.g., digital), and may apply the driving signal in an analog form to the liquid lens **280** through pulse-width modulation (PWM) control. Accordingly, finally, the controller **240** may control the curvature of the liquid lens **280**.

The lens driving unit **250** may selectively provide a voltage level supplied from the power supply unit **260**, which will be described below, to each terminal of the liquid lens **280**. In an embodiment, the lens driving unit **250** may include a switching unit. Here, the switching unit may include a circuit component called an H-bridge.

In addition, a high voltage output from a voltage booster may be applied as a power supply voltage of the switching unit. Accordingly, the switching unit may selectively supply the applied power supply voltage and a ground voltage to both ends of the liquid lens **280**.

Further, as described above, the liquid lens **280** includes four first electrodes including four electrode sectors for driving, a first connection substrate, one second electrode, and a second connection substrate. Both ends of the liquid lens **280** may be any one of the plurality of first electrodes and the second electrode. In addition, both ends of the liquid lens **280** may be one of the four electrode sectors of the four first electrodes and one electrode sector of the second electrode.

Accordingly, a pulse-type voltage having a predetermined width may be applied to each electrode sector of the liquid lens **280**. In addition, the voltage may be a difference between voltages applied to the first and second electrodes, and may be applied to the liquid lens **280**.

The power supply unit **260** may apply a separate power to the lens driving unit **250**. The power supply unit **260** may include the voltage booster that increases a voltage level. In addition, the aforementioned lens driving unit **250** may selectively output the increased voltage to each terminal of the liquid lens **280**.

As described above, the controller **240** may control a phase of the driving signal supplied to the lens driving unit **250**, that is, a phase of a pulse voltage applied to the common electrode and the individual electrode so that the lens driving unit **250** may generate an analog-type driving voltage corresponding thereto. In other words, the controller **240** may control a voltage applied to each of the first electrode and the second electrode. Hereinafter, a description will be made on the basis of a driving signal in the form of voltage.

In addition, the control circuit **210** may further include a connector (not shown) that performs a communication or interface function of the control circuit **210**. For example, the connector may perform communication protocol conversion for communication between the control circuit **210**, which uses an I2C communication method, and the lens assembly **220**, which uses a mobile industry processor interface (MIPI) communication method. In addition, the connector may receive power from an external device (e.g., a battery) and supply power required for the operation of each of the controller **240** and the lens assembly **220**. In this case, the connector may correspond to the connector **153** shown in FIG. 2.

Referring to FIG. 10, a camera module according to the modified example may include the components described in FIG. 9. That is, the same content may be applied to the components described in FIG. 9, and components other than those described in FIG. 9 will be described below.

First, the control circuit **210** may further include an equalizer **270**, a sensor unit **291**, a converter **292**, and a memory **290**. However, it should be understood that positions of the sensor unit **291** and the converter **292** are not limited to such positions and the sensor unit **291** and the converter **292** may be provided at any position in the camera module.

First, when a calculated curvature is greater than a target curvature, the equalizer **270** may control a duty of a pulse width to be increased or control a delay, which is a time difference between a plurality of pulses applied to the liquid lens **280**, to be increased on the basis of a calculated

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curvature error (D). Accordingly, the camera module may quickly and accurately change the curvature of the liquid lens **280**.

The equalizer **270** may receive focus information AF from the image processor **31** and shake information OIS from the gyro sensor **230**, and determine the target curvature on the basis of the focus information AF and the inclination information (or shake information) OIS.

Here, an update cycle of the determined target curvature may be longer than an update cycle of the calculated curvature on the basis of the detected capacitance of the liquid lens **280**. However, the present invention is not limited thereto.

In addition, since the update cycle of the calculated curvature is shorter than the update cycle of the target curvature, the curvature of the liquid lens **280** may be varied to be quickly changed to a desired curvature.

The memory **290** may be positioned separately from the control circuit **210**. In addition, as described above, in order to form the target curvature, the control circuit **210** may output a changed voltage signal on the basis of the detected temperature, to the liquid lens **280** from the lens driving unit **250**. In addition, the control circuit **210** may vary a delay, which is a time difference of a plurality of pulses applied to the liquid lens **280**, according to the detected temperature.

Meanwhile, the memory **290** may store temperature information according to each diopter, delay information, information on a delay that is a time difference between a plurality of pulses (voltage signals) applied to the liquid lens **280**, gain information, and offset information. For example, the memory **290** may store temperature information according to a first reference diopter (e.g., a 0 diopter), delay information, information on a delay that is a time difference between a plurality of pulses applied to the liquid lens **280**, gain information, and offset information.

Further, the control circuit **210** may quickly and accurately vary the curvature of the lens using the detected temperature information by varying a delay, which is a time difference of a plurality of pulses applied to the liquid lens **280**, according to the detected temperature.

That is, the control circuit **210** may vary the curvature of the lens quickly and accurately using the detected temperature information by varying a delay, which is a time difference of a plurality of pulses applied to the liquid lens **280**, according to the information stored in the memory **290** and the detected temperature.

Further, the control circuit **210** may output a compensation signal for compensating for an offset of the signal sensed by the sensor unit **291** according to the detected temperature, thereby accurately detecting the curvature of the lens.

Further, the converter **292** may convert a signal related to the capacitance detected by the sensor unit **291** into a digital signal.

In addition, the camera module **200** may further include conductive lines CA1 and CA2 and a switching element SWL. First, the plurality of conductive lines CA1 and CA2 may supply an electrical signal from the lens driving unit **250** to each of the electrodes (the common electrode and the plurality of electrodes) in the liquid lens **280**. In addition, the switching element SWL may be disposed between one (e.g., CA2) of the plurality of conductive lines and the sensor unit **291**.

In the present specification, when referring to the drawings, by way of example, the conductive line CA2 may apply an electrical signal to any one of the plurality of electrodes in the liquid lens **280**, and the switching element SWL may

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be disposed between the conductive line CA2 and the sensor unit **291**. In this case, a contact point between the conductive line CA2 and one end of the switching element SWL or the liquid lens **280** may be referred to as node A.

Meanwhile, in the present invention, an electrical signal may be applied to each of the electrodes (the common electrode and the plurality of electrodes) in the liquid lens **280** through the plurality of conductive lines CA1 and CA2 to detect the curvature of the liquid lens **280**. Accordingly, a curvature may be formed in the liquid in the liquid lens as described above.

For example, the switching element SWL may be turned on during a first period. At this point, when an electrical signal is applied to the electrodes in the liquid lens **280** in a state in which the switching element SWL is turned on and is thus electrically connected to the sensor unit **291**, a curvature may be formed in the liquid lens **280**. In addition, an electrical signal corresponding to the formed curvature may be supplied to the sensor unit **291** via the switching element SWL.

Accordingly, the sensor unit **291** may detect a size of an area or a change in the area of a boundary region in the liquid lens **280** of the liquid lens **280** on the basis of the electrical signal from the liquid lens **280** during the ON period of the switching element SWL. In an embodiment, the sensor unit **291** may detect a capacitance of the boundary region.

In addition, during a second period after the first period, the switching element SWL may be turned off, and the electrical signal may be continuously applied to the electrodes in the liquid lens **280**. Accordingly, a curvature may be formed in the boundary region in the liquid lens.

In addition, during a third period after the second period, the switching element SWL may be turned off, and no electrical signal or a low-level electrical signal may be applied to the electrodes in the liquid lens **280**.

In addition, during a fourth period after the third period, the switching element SWL may be turned on. At this point, the switching element SWL may be turned on and thus may be in a state of being electrically connected to the sensor unit **291**. In addition, when an electrical signal is applied to the electrodes in the liquid lens **280**, a curvature may be formed in the liquid lens **280**, and an electrical signal corresponding to the formed curvature may be supplied to the sensor unit **291** via the switching element SWL.

Meanwhile, when the curvature calculated on the basis of the capacitance detected during the first period is less than the target curvature, the control circuit **210** may control a pulse width of the driving signal supplied to the lens driving unit **250** to be increased so that the curvature reaches the target curvature.

Thus, the time difference between the pulses applied to the plurality of individual electrodes E1 and the common electrode E2 may be increased, and accordingly, the curvature formed in the boundary region may be changed.

In addition, since the sensor unit **291** detects the curvature of the boundary region, the camera module may quickly and accurately identify the curvature of the liquid lens **280**.

In addition, the lens driving unit **250**, the sensor unit **291**, the control circuit **210**, the power supply unit **260**, the converter **292**, and the switching element SWL may be implemented by a single chip as a system on chip (SOC). However, the present invention is not limited thereto.

Referring to FIG. 11, a timing diagram for the common electrode E2, the individual electrode E1, and the switching element SWL in the camera module is illustrated.

Specifically, during a period Dt1 between a time point of T1 and a time point of T3, the switching element SWL is

turned on. In this case, in order to detect the capacitance of the boundary region through the sensor unit **291**, a curvature may be formed in the liquid lens **280** during the period **Dt1** between the time point of **T1** and the time point of **T3**.

In addition, it is illustrated that, for the accuracy and stability of detection in the sensor unit **291**, in the present invention, a pulse is applied to one of the common electrode **E2** and the plurality of individual electrodes **E1** in the liquid lens **280** during the period **Dt1** between the time point of **T1** and the time point of **T3**.

In an embodiment, a pulse having a pulse width of **Dt2** may be applied to the common electrode **E2** at a time point of **T2**. Accordingly, a curvature may be formed in the liquid lens **280** after the time point of **T2**.

In addition, as described above, the sensor unit **291** may detect capacitances corresponding to the size of the area or the change in the area of the boundary region during a period between the time point of **T2** and the time point of **T3** in the period **Dt1** between the time point of **T1** and the time point of **T3**. Alternatively, during the period between the time point of **T2** and the time point of **T3**, the sensor unit **291** may detect a potential difference or a current between the first liquid and the electrodes corresponding to the size of the area or the change in the area of the boundary region in the liquid lens **280**.

In addition, a pulse having a pulse width of **Dt3** may be applied to the individual electrode **E1** at a time point of **T4**. That is, a high-level voltage may be applied to the common electrode **E2** at the time point of **T2**, and a high-level voltage may be applied to the individual electrode **E1** at the time point of **T4**.

Accordingly, the curvature in the liquid lens **280** may be varied due to a time difference **DFF1** between the pulse applied to the common electrode **E2** and the pulse applied to the individual electrode **E1**.

For example, as the time difference **Dff1** between the pulses increases, the area of the boundary region may increase. In other words, the capacitance may be increased and the curvature may be increased.

In addition, it is possible to maintain or change the curvature in the liquid lens **280** by varying a time difference **DFF2** between the pulses after the time difference **DFF1** between the pulses.

In addition, it should be understood that, as will be described below in the camera module according to the embodiment, the controller may perform a control of varying a weight and performing compensation according to a weight ratio simultaneously with or separately from the compensation performed to obtain the above-described target curvature. For example, the function of the controller, which will be described below, may be achieved independent of the compensation, which is performed to obtain the target curvature, as in FIG. 9.

FIGS. 12A and 12B are views for describing a driving of a camera module in response to shaking, FIGS. 13A and 13B are views for describing a case in which different driving signals are applied depending on weights, FIG. 13C is a view illustrating a change in an interface according to an application of driving signals having different weights, and FIG. 14 is a view illustrating a wavefront error (WFE) according to weights.

First, referring to FIGS. 12A and 12B, the liquid lens unit **140** may be shaken due to a shaking of a user's hand or the like. Accordingly, focusing on the image sensor may not be performed accurately through the liquid lens unit **140**.

Further, the gyro sensor in the camera module may detect the shake information described above. In an embodiment,

the gyro sensor may output inclination information. The inclination information may include an inclination (angle) of the camera module or liquid lens with respect to a reference position. For example, the gyro sensor may output information on an angular velocity, but may output information on an inclination through the angular velocity, which will now be described as inclination information.

Further, since the degree of the shaking of the user's hand is varied, the inclination (°) obtained through the gyro sensor is also varied, but the following description of the present specification is made on the premise that a change (from a to -a) in the inclination for a predetermined time (or a cycle **T**) is constant. In addition, it will be described that the driving signal also has a cycle **T** corresponding to the cycle **T**. In addition, it will be described that the gyro sensor detects the inclination of two axes and the two axes have a first axis (**X**) and a second axis (**Y**). In addition, the first axis (**X**-axis) and the second axis (**Y**-axis) may each correspond to any one of a yaw axis and a pitch axis, but the present invention is not limited thereto.

Referring to FIG. 13A, the liquid lens unit **140** may vary the interface **BO** of the liquid according to the shaking. That is, when the liquid lens unit **140** is inclined by as much as a first angle θ in response to the shaking of the camera module, the interface may be changed to compensate for the shaking. In this case, the first angle θ may be an angle with respect to a vertical axis **OXL** of an optical axis **PXL** before the shaking. In addition, before the shaking may mean a time point at which the focus information **AF** is received from an image processing unit, but the present invention is not limited thereto.

For example, in FIG. 13A, a right side of the interface **BO** may be convex to compensate for the shaking, and it will be described below based on an interface **BOP** before the interface is changed in response to the shaking. For convenience of description, the interface **BOP** before the interface is changed in response to the shaking will be described as a first interface, and the interface **BO** changed according to the shaking will be described as a second interface. In addition, as described above, the first interface **BOP** and the second interface **BO** are boundary regions between the first liquid **LQ1** and the second liquid **LQ2**.

As it is deformed from the first interface **BOP** to the second interface **BO**, an interface angle may also be changed. The interface angle may be an angle formed between the interface and the first plate **147** or the insulating layer **148**.

In an embodiment, the first interface **BOP** may form a first interface angle $\theta 1$ and a second interface angle $\theta 2$. In addition, the second interface **BO** may form a third interface angle $\theta 1'$ and a fourth interface angle $\theta 2'$.

In addition, as it is deformed from the first interface **BOP** to the second interface **BO**, the first interface angle $\theta 1$ may be changed into the third interface angle $\theta 3$, and the second interface angle $\theta 2$ may be changed into the fourth interface angle $\theta 4$.

In addition, as described above, since the liquid lens unit **140** is inclined by as much as the first angle θ , when the optical axis **PXL** before shaking is used as a reference, an inclination in one region (hereinafter described as a first corresponding region **S1** corresponding to the first-first electrode **E11**) and an inclination in another region (hereinafter described as a second corresponding region **S2** corresponding to the first-second electrode **E12**) opposite to the one region may be complementary. For example, when an angular velocity in the first corresponding region **S1** increases (when the angle is positive), an angular velocity in

the second corresponding region S2 may decrease (the angle is negative). In other words, when the angular velocity of the liquid lens unit 140 increases, the inclination of the liquid lens in the corresponding region may become positive, and when the angular velocity thereof decreases, the inclination of the liquid lens in the corresponding region may become negative.

Here, the first corresponding region is a region in which the interface is changed corresponding to any one of the individual electrodes as shown in FIG. 7B. For example, when the individual electrodes are four as described above, first to fourth corresponding regions may be present.

That is, the gyro sensor may output inclination information in which the angular velocity increases in the first corresponding region S1 of the liquid lens and decreases in the second corresponding region S2 due to the shaking. In addition, it may be changed from the first interface BOP to the second interface BO due to the shaking. That is, the interface at the first corresponding region S1 may move upward (b), but the interface at the second corresponding region S2 may move downward (a).

Further, when the third interface angle θ_3 is an angle decreased from the first interface angle θ_1 , the fourth interface angle θ_4 may be an angle increased from the second interface angle θ_2 . In contrast, when the third interface angle θ_3 is an angle increased from the first interface angle θ_1 , the fourth interface angle θ_4 may be an angle decreased from the second interface angle θ_2 .

The controller according to the embodiment may apply different weights to the driving signal according to the increase or decrease of the angular velocity of the liquid lens which is obtained on the basis of the inclination information. Specifically, the controller may apply a first weight to the driving signal when the interface moves downward (or when the interface angle increases) due to the shake information in which the angle of the liquid lens decreases. In addition, the controller may apply a second weight to the driving signal when the interface moves upward (or when the interface angle increases) due to the shake information in which the angle of the liquid lens increases. In addition, the first weight may be less than the second weight.

In other words, when the angular velocity of the inclination information of the liquid lens decreases, the controller according to the embodiment may apply a weight that allows the driving signal to be increased. Alternatively, when the angular velocity of the inclination information of the liquid lens increases, the controller may apply a weight that allows the driving signal to be decreased.

Accordingly, the controller according to the embodiment may inhibit the interface from moving upward in the first corresponding region after moving downward in the second corresponding region. That is, the controller causes the interface to move with a similar reaction speed as the angular velocity increases or decreases. As a result, the camera module according to the embodiment may minimize a wavefront error (WFE) by controlling the movement (upward or downward movement) of the interface to equally occur. In addition, when the interface moves upward, the controller may rapidly perform OIS by reducing the reaction speed.

Further, the first weight and the second weight may be applied to the above-described time difference between the pulse applied to the common electrode E2 and the pulse applied to the individual electrode E1.

In an embodiment, the controller may vary the time difference between the pulses of the driving signal, which are applied to the liquid lens, according to the first weight

and the second weight. That is, the first weight and the second weight may be applied to increase or decrease the above-described time difference between the pulse applied to the common electrode E2 and the pulse applied to the individual electrode E1.

In addition, a first delay corresponding to the first weight and a second delay corresponding to the second weight may be present in the time difference. In addition, when the same reference driving signal is applied, the first delay may be greater than the second delay. Accordingly, the wavefront error (WFE) may be minimized at the interface, and the reaction speed of the interface according to the driving signal may be improved. However, the present invention is not limited thereto, and it should be understood that the second interface BO may be raised or lowered in both the first corresponding region S1 and the second corresponding region S2.

For example, when the interface is raised in both the first corresponding region S1 and the second corresponding region S2, the interface angle decreases in both regions, and thus the second weight may be applied. In addition, when the interface is lowered in both the first corresponding region S1 and the second corresponding region S2, the interface angle increases in both regions, and thus the first weight may be applied.

Referring to FIG. 13B, it can be seen that different weights may be applied to the driving signal in response to the change (the increase or decrease of the angular velocity) in the angle shown in FIG. 12B.

Specifically, the controller may control the pulse width of the driving signal, which is a voltage code in a data form (e.g., digital) as described above, and provide an analog-type voltage illustrated in FIG. 13B to the individual electrodes of the liquid lens via the lens driving unit 250. Accordingly, the controller may accurately and rapidly control the curvature of the liquid lens in response to shaking, and may also minimize a wavefront error (WFE) according to the interface.

When there is no shaking, the controller may output a driving signal (hereinafter, referred to as a reference driving signal) corresponding to a signal to which AF is applied, that is, a signal received from an external system. In this case, the reference driving signal may have a magnitude of v_1 .

However, when there is shaking, the controller may output a driving signal that varies in response to the shaking. In other words, the controller according to the embodiment may change the reference driving signal in order to compensate for shaking, and depending on the increase or decrease of the angle, the controller may apply different weights for the change of the reference driving signal.

For example, when a first-fourth electrode E14 compensates for the first axis (X^+) in the liquid lens, the first-fourth electrode E14 may provide a driving signal having a voltage less than the magnitude v_1 of the reference driving signal until TO at which the angular velocity increases. In this case, the driving signal may have a maximum value of v_3 .

Alternatively, when the first-second electrode E12 compensates for the second axis (Y^+) in the liquid lens, the first-second electrode E12 may provide a driving signal having a voltage greater than the magnitude v_1 of the reference driving signal until TO at which the angular velocity decreases. In this case, the driving signal may have a maximum value of v_2 .

In this case, a voltage difference K1 between the minimum value v_3 and the magnitude v_1 of the reference driving signal may be greater than a voltage difference K2 between the maximum value v_2 and the magnitude v_1 of the refer-

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ence driving signal. This may be a result due to the difference in the above-described weights. In addition, when there is no shaking, the minimum value and the maximum value may be the same as the reference driving signal.

Referring to FIG. 13C, a variation of the interface of the liquid lens according to the embodiment and the diopter are illustrated. In addition, after the inclination information is output from the gyro sensor and then a predetermined time Δt_d has passed, the interface angle may be changed. It should be understood that this is due to a processing speed required for the controller to receive the inclination information, use the inclination information to apply weights to the driving signal, and then apply the driving signal to the liquid lens.

In addition, it can be seen that the wavefront error (WFE) is minimized since a change in the interface angle with respect to each of the X^+ and Y^+ (axes) is made without a time difference during a cycle T .

In addition, the diopter indicates a change in focus (or focusing) averaged in the individual electrodes E11 to E14. This will be described below.

Referring to FIG. 14, a wavefront error (WFE) according to a weight ratio (the second weight to the first weight) in the controller according to the embodiment is illustrated.

First, the wavefront error (WFE) was measured while changing the weight ratio from 1:0.6 to 1:2.3. In this case, it can be seen that when the weight ratio is in a range of 1:1.1 to 1:1.5, the wavefront error is less than that in the weight ratio of 1:1. That is, when the weight ratio is less than 1:1.1 or greater than 1:1.5, the wavefront error (WFE) increases, and thus, there is a limit that the resolution of the camera module is reduced. In addition, when the weight ratio is greater than 1:1.5, the interface moves upward in a region in which the angular velocity increases, and thus there is a limit that the reaction speed is further delayed.

FIG. 15 is a view illustrating a change in diopter for a half cycle of the driving signal, FIG. 16 is a view for describing diopter defocusing, FIG. 17 is a view illustrating a variation of the driving signal during compensation, FIG. 18 is a view for describing compensation according to a weight difference, and FIG. 19 is a view illustrating an average voltage value according to compensation.

First, the controller according to the embodiment may compensate for a difference between the first weight and the second weight in response to a half cycle of the driving signal.

Referring to FIG. 15, when a time difference Δt_d due to signal processing is excluded, the diopter is increased or decreased based on an average diopter AVEd during the cycle T as described in FIG. 13C. Here, the average diopter AVEd is an average of voltages of the driving signal (or voltages applied to the individual electrodes) when AF is followed without performing OIS. Further, since the controller according to the embodiment has the first weight greater than the second weight, as described above, a decreasing rate of the voltage is greater than an increasing rate of the voltage. Accordingly, a change range of the diopter with respect to the average diopter AVEd may be increased.

Referring to FIG. 16, the controller according to the embodiment may increase a change range ΔdI of the diopter by varying the first weight and the second weight. In addition, the controller may provide a center diopter dm with a value smaller than that (zero (0) in the drawing) of the diopter through the AF only by varying the weight.

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As a result, as the center diopter dm is smaller than 0 and the change range ΔdI increases due to the variation of the weight, the occurrence of defocusing may be increased.

Accordingly, the controller according to the embodiment may apply a driving signal according to Equation 1 below.

$$\begin{bmatrix} V_x^+ \\ V_x^- \\ V_y^+ \\ V_y^- \end{bmatrix} = V_0 \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} +G_{tilt}(g1 \text{ or } g2) & 0 & \varepsilon \\ -G_{tilt}(g1 \text{ or } g2) & 0 & \varepsilon \\ 0 & +G_{tilt}(g1 \text{ or } g2) & \varepsilon \\ 0 & -G_{tilt}(g1 \text{ or } g2) & \varepsilon \end{bmatrix} \quad \text{Equation 1}$$

$$\begin{bmatrix} \theta_x \\ \theta_y \\ \max_{\frac{\pi}{4} \leq \theta \leq \frac{3\pi}{4}} \frac{1}{2} [|\theta_x - \theta_y| + |\theta_x + \theta_y|] - (|\theta_x - \theta_y| + |\theta_x + \theta_y|) \end{bmatrix}$$

where V_x^+ , V_x^- , V_y^+ , and V_y^- are each a voltage of the driving signal applied to each individual electrode, V_0 is a voltage applied to the individual electrode to set a focus according to AF or a voltage before the compensation for shaking, G_{tilt} is a linear correlation coefficient between an optical inclination of the liquid lens and a voltage offset between opposite electrodes (e.g., X^+ and X^- or Y^+ and Y^-), $g1$ is the second weight, $g2$ is the first weight, and F is a focus compensation parameter according to the first and second weights in the controller. In addition, θ_x is an x-axis angle, and θ_y is the y-axis angle.

Further, the x-axis angle and the y-axis angle are obtained by Equation 2 below.

$$\begin{pmatrix} \theta_x \\ \theta_y \end{pmatrix} = \theta_{opt} \cdot \begin{pmatrix} \cos(\Phi_0) \\ \sin(\Phi_0) \end{pmatrix} \quad \text{Equation 2}$$

where θ_{opt} is an optical inclination induced by a geometric inclination of the interface of the liquid lens, and is represented by Snell-Descartes' law of refraction. In addition, Φ_0 is an azimuth angle and is an angle corresponding to a non-uniform voltage other than V_0 , which affects the curvature of the interface, and may be changed according to a shape of the interface when the diopter due to AF is 0 (for example, the interface is parallel to the connecting substrate). In the present specification, since the interface is flat at 0 diopter, Φ_0 is 0° .

The controller according to the embodiment may compensate for the difference between the first weight and the second weight in response to a half cycle of the driving signal.

Specifically, according to the embodiment, due to a ratio between the second weight and the first weight, a decreasing rate of the driving signal may be further increased at every half cycle.

Referring to FIG. 15 again, it can be seen that the cycle T and amplitude of the diopter change to correspond to a change in $|\theta_x - \theta_y| + |\theta_x + \theta_y|$. That is, the change in diopter and the change in $|\theta_x - \theta_y| + |\theta_x + \theta_y|$ may be identically performed according to time.

However, as shown in FIG. 13A, since an angular velocity variation occurs every half cycle, the controller according to the embodiment may control the driving signal to compensate for the change in diopter at every half cycle for $|\theta_x - \theta_y| + |\theta_x + \theta_y|$.

More specifically, as in Equation 1 described above, the controller may compensate for the change range of the

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dioptr and the center dioptr, which are increased according to the weight ratio between the first weight and the second weight, through $\max_{half\ cycle} [|\theta_x - \theta_y| + |\theta_x + \theta_y|] - (|\theta_x - \theta_y| + |\theta_x + \theta_y|)$.

In an embodiment, since the first weight is greater than the second weight, the controller may apply $\max_{half\ cycle} [|\theta_x - \theta_y| + |\theta_x + \theta_y|] - (|\theta_x - \theta_y| + |\theta_x + \theta_y|)$ to each individual electrode to increase the magnitude of the driving signal (corresponding to the voltage) applied to the individual electrode. As a result, the controller may increase an average voltage difference of the driving signal for each cycle.

Referring to FIG. 17, it can be seen that different weights may be applied to the driving signal in response to the change (increase or decrease of the angular velocity) in the angle shown in FIG. 12B.

As described above, the controller may control the pulse width of the driving signal, which is a voltage code in a data form (e.g., digital) as described above, and provide an analog-type voltage illustrated in FIG. 13B to the individual electrodes of the liquid lens via the lens driving unit 250. Accordingly, the controller may accurately and rapidly control the curvature of the liquid lens in response to shaking, and may also minimize a wavefront error (WFE) according to the interface.

In addition, when there is no shaking, the controller may output a driving signal (hereinafter, referred to as a reference driving signal) corresponding to a signal to which AF is applied, that is, a signal received from an external system. In this case, the reference driving signal may have a magnitude of $v1$.

However, when there is shaking, the controller may output a driving signal that varies in response to the shaking. In other words, the controller according to the embodiment may change the reference driving signal in order to compensate for shaking, and depending on the increase or decrease of the angle, the controller may apply different weights for the change of the reference driving signal.

For example, when the first-fourth electrode E14 compensates for the first axis (X^+) in the liquid lens, the first-fourth electrode E14 may provide a driving signal having a voltage less than the magnitude $v1$ of the reference driving signal until TO at which the angular velocity increases. In this case, the driving signal may have a maximum value of $v3$.

Alternatively, when the first-second electrode E12 compensates for the second axis (Y^+) in the liquid lens, the first-second electrode E12 may provide a driving signal having a voltage greater than the magnitude $v1$ of the reference driving signal until TO at which the angular velocity decreases. In this case, the driving signal may have a maximum value of $v2$.

Further, likewise, a voltage difference K1 between the minimum value $v3$ and the magnitude $v1$ of the reference driving signal may be greater than a voltage difference K2 between the maximum value $v2$ and the magnitude $v1$ of the reference driving signal. This may be a result due to the difference in the above-described weights. In addition, when there is no shaking, the minimum value and the maximum value may be the same as the reference driving signal.

In addition, referring to FIG. 18, when a first-third electrode E13 compensates for a third axis (X^-) and the first-first electrode E11 compensates for a fourth axis (Y^-), the controller may apply a driving signal having a voltage greater than the magnitude $v1$ of the reference driving signal to the first-third electrode E13 and the first-first electrode E11 at every half cycle, which is a result of applying $\max_{half\ cycle} [|\theta_x - \theta_y| + |\theta_x + \theta_y|] - (|\theta_x - \theta_y| + |\theta_x + \theta_y|)$ to each of

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the individual electrodes E11 to E14 in response to Equation 1 described above (in addition, $\max_{half\ cycle} [|\theta_x - \theta_y| + |\theta_x + \theta_y|] - (|\theta_x - \theta_y| + |\theta_x + \theta_y|)$ is applied to the first-second electrode E12 and the first-fourth electrode E14 at every half cycle).

In addition, since $|\theta_x - \theta_y| + |\theta_x + \theta_y|$ and the size of the change in the dioptr are different, the controller according to the embodiment may perform accurate compensation by controlling a focus compensation parameter F according to the first and second weights. That is, the magnitude of the driving signal of the individual electrode corresponding to the weight ratio may be compensated for by applying the focus compensation parameter ϵ to $\max_{half\ cycle} [|\theta_x - \theta_y| + |\theta_x + \theta_y|] - (|\theta_x - \theta_y| + |\theta_x + \theta_y|)$.

Since the controller according to the embodiment controls the weight ratio between the first weight and the second weight to be in a range of 1:1.1 to 1:1.5, the focus compensation parameter F according to the first and second weights may be 1 or less. In addition, the focus compensation parameter F may be in a range of 0.26 to 0.33. Accordingly, the controller according to the embodiment reduces the wavefront error (WFE) to improve the resolution and improve the reaction speed of the interface, and simultaneously, minimize defocusing. For example, by compensating for the difference between the first weight and the second weight at every half cycle through the controller, a focus change may be adjusted within a range that is not recognized by a person.

Referring to FIG. 19, as the controller varies the weight and compensates to correspond to the weight ratio (e.g., apply focus compensation parameters), a center dioptr dm' may move closer to zero as compared to the center dioptr dm (see FIG. 16) in the case in which the compensation is not performed. That is, the controller may increase the average voltage difference of the driving signal for each average cycle through the compensation. In addition, a variation range $\Delta d2$ of the dioptr generated by the variation of the weight may be reduced to suppress the defocusing.

Another controller according to a modified example may apply different first and second weights for a predetermined time and equally reflect a third weight after the predetermined time. For example, the first weight and the second weight may be changed to one. Accordingly, the predetermined time may be greater than the cycle of the driving signal. Accordingly, the wavefront error (WFE) may be reduced to improve the resolution as described above, and the occurrence of defocusing may be suppressed through the third weight.

The invention claimed is:

1. A camera module comprising:

a lens assembly including a liquid lens unit having a first liquid and a second liquid that form an interface;
a gyro sensor configured to output inclination information; and

a controller configured to adjust the interface by applying a driving signal to the liquid lens unit based on focus information and the inclination information,
wherein the controller is configured to apply different weights to the driving signal according to an angular velocity of the liquid lens unit obtained based on the inclination information,

wherein the controller is configured to apply a first weight to the driving signal when the angular velocity of the liquid lens unit increases, and apply a second weight to the driving signal when the angular velocity of the liquid lens unit decreases,

wherein the first weight is greater than the second weight,

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wherein the controller is configured to compensate for a difference between the first weight and the second weight in response to a half cycle of the driving signal, and

wherein the driving signal is a signal applied to each individual electrode coupled with the liquid lens unit that is configured to change a curvature of the interface.

2. The camera module of claim 1, wherein the controller is configured to vary a time difference between a plurality of pulses applied to the liquid lens unit according to the first weight and the second weight.

3. The camera module of claim 2, wherein the time difference includes a first delay corresponding to the first weight and a second delay corresponding to the second weight, and

wherein the first delay is greater than the second delay.

4. The camera module of claim 1, wherein a ratio of the second weight and the first weight is in a range of 1:1.1 to 1:1.5.

5. The camera module of claim 1, wherein the controller is configured to increase an average voltage difference of the driving signal for each cycle during compensation.

6. The camera module of claim 5, wherein the controller is driven according to Equation 1 below,

$$\begin{bmatrix} V_x^+ \\ V_x^- \\ V_y^+ \\ V_y^- \end{bmatrix} = V_0 \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} +G_{nlt}(g1 \text{ or } g2) & 0 & \varepsilon \\ -G_{nlt}(g1 \text{ or } g2) & 0 & \varepsilon \\ 0 & +G_{nlt}(g1 \text{ or } g2) & \varepsilon \\ 0 & -G_{nlt}(g1 \text{ or } g2) & \varepsilon \end{bmatrix} \cdot \begin{bmatrix} \theta_x \\ \theta_y \\ \max_{\theta_x, \theta_y} [|\theta_x - \theta_y| + |\theta_x + \theta_y|] - (|\theta_x - \theta_y| + |\theta_x + \theta_y|) \end{bmatrix} \quad \text{Equation 1}$$

(where V_x^+ , V_x^- , V_y^+ , and V_y^- each is a voltage of the driving signal applied to each individual electrode, V_0 is a voltage applied to each individual electrode to set

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a focus according to auto-focusing (AF) or a voltage before a compensation for shaking, G_{nlt} is a linear correlation coefficient between an optical inclination of the liquid lens unit and a voltage offset between opposite electrodes (including X^+ and X^- or Y^+ and Y^- , $g1$ is the second weight, $g2$ is the first weight, and E is a focus compensation parameter according to the first and second weights provided by the controller; θ_x is an x-axis angle, and θ_y is a y-axis angle).

7. The camera module of claim 1, comprising a sensor unit configured to detect interface information corresponding to a curvature of the interface.

8. The camera module of claim 7, wherein the sensor unit detects a capacitance corresponding to a size of the interface or a change in an area of the interface.

9. The camera module of claim 1, comprising an image sensor configured to receive light passing through the lens assembly,

wherein the focus information is based on image information output from the image sensor.

10. The camera module of claim 1, wherein the controller is configured to change the weights equally after a predetermined time.

11. The camera module of claim 1, wherein the liquid lens unit includes:

a plate including a cavity in which a conductive liquid and a non-conductive liquid are disposed while forming an interface;

a plurality of individual terminals disposed below the plate; and

a common terminal disposed above the plate.

12. The camera module of claim 11, wherein the driving signal is applied to the plurality of individual terminals.

13. The camera module of claim 11, wherein the driving signal includes a signal applied to at least one of the plurality of individual terminals and the common terminal for a predetermined cycle.

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